# Channel Estimation for Massive MIMO-OFDM With Generalized Phase Shift Pilots

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***Abstract*—We investigate a non-orthogonal pilot design for massive multiple-input multiple-output (MIMO) channel estimation with orthogonal frequency division multiplexing (OFDM) modulation. Leveraging two-dimensional (2D) beam based channel model, we formulate a signal model for channel estimation with a comb-type pilot structure. Then, we propose generalized phase shift pilots (GPSPs) and reveal that both GPSPs and their discrete Fourier transform (DFT) have constant modulus, and the DFT of GPSPs also has optimal autocorrelation property and beneficial crosscorrelation property. Subsequently, we prove that the inter-user interference with GPSPs is negligible when the weight sequence length of GPSPs is large enough or when channels can be differentiated in the angle domain, validating the feasibility of GPSPs, especially when the statistical channel state information (CSI) is unavailable. Further, by leveraging the correlation properties of GPSPs, an efficient implementation for GAMP based channel estimation is provided. Simulation results indicate that GPSPs enable low-complexity channel estimation while ensuring satisfactory performance.**

***Index Terms*—Massive MIMO-OFDM, beam based channel model, generalized phase shift pilots, channel estimation.**

## I. Introduction

MASSIVE multiple-input multiple-output orthogonal frequency division multiplexing (MIMO-OFDM) [1], [2], [3], [4] is a fundamental enabling technology for the fifth generation (5G) mobile communication, and will play a key role for the sixth generation (6G) mobile communication with increased system configuration in antenna and user terminal (UT) numbers. In massive MIMO systems, a large number of antennas are deployed at the base station (BS) to simultaneously serve a number of UTs on the same time and frequency resources. Therefore, massive MIMO can effectively exploit spatial wireless resources and then significantly enhance spectral efficiency [5], [6], [7].

The accurate acquisition of channel state information (CSI) is essential for harnessing the potential capacity gains provided by massive MIMO-OFDM. Although conventional orthogonal pilots [8], [9] can eliminate pilot interference in the same cell, they also cause excessive overhead with the increasing numbers of antennas and UTs in massive MIMO-OFDM systems. On the other hand, non-orthogonal pilots enable systems to support a greater number of users without incurring additional overhead, and numerous relevant studies have been conducted. The limited angle spread in massive MIMO channels enables the reuse of the same pilots for UTs with non-overlapping angle domain channels [10], [11], [12]. Assuming the knowledge of visible regions is known, a two-stage graph partitioning is proposed to design non-orthogonal pilots, which reduces the pilot overhead without compromising sum rate performance [13]. Adjustable phase shift pilots (APSPs) [14] with a pilot phase shift scheduling algorithm can fully utilize wireless resources in both space and frequency dimensions, thereby serving a greater number of UTs than phase shift orthogonal pilots (PSOPs) [9], [15], while similar channel estimation performance can be maintained. Therefore, with increased UTs in massive MIMO-OFDM systems, non-orthogonal pilots can fully harness their strengths, which makes them more critical. Nevertheless, these pilots require the statistical CSI, including the space-frequency (SF) statistical CSI or at least the spatial statistical CSI. Since obtaining statistical CSI is relatively hard before channel estimation [16], the application of these pilots might be limited in the practical systems. Thus, a non-orthogonal pilot that can provide satisfactory performance without the statistical CSI is in great demand.

The performance of channel estimation also depends on the channel model and the channel estimation methods, both of which have been extensively studied over the past decades. Conventional channel estimation methods, such as least square (LS) and minimum mean square error (MMSE) methods, fail to operate well in massive MIMO-OFDM systems because of the unaffordable computational complexity caused by large-dimensional matrix inversion. Due to the limited number of local scatterers and the small angle spread in practical massive MIMO systems, the angle domain channels exhibit sparsity [17]. Besides, the sparse multipath structure of wire-

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less channel and the limited delay spread suggest the channel sparsity in the delay domain [18]. To be specific, beginning from a physical model of wireless channel, the SF domain channel can be transformed to a two-dimensional (2D) beam domain channel [19], [20], [21]. The double sparsity of the 2D beam domain channel allows a reduction in the number of parameters to be estimated. Thus, channel estimation can be effectively realized by compressed sensing methods. Specifically, orthogonal matching pursuit (OMP) method [22] and its variants [23] utilize greedy schemes to iteratively select the sparse support and then reconstruct the channel with these supports by LS method. Furthermore, generalized approximate message passing (GAMP) algorithm [24] utilizes approximations in the factor graph, enabling an efficient channel recovery only involving matrix multiplications and element-wise operations, while Bayes-optimal performance can be approached [25].

In this paper, we propose generalized phase shift pilots (GPSPs) for massive MIMO-OFDM channel estimation, which generalize the phase shift pilots (PSPs) by linearly combining a set of PSPs with different weights. Different root sequences for the PSPs are assigned to different cells, while different weight sequences are used to distinguish different UTs in the same cell. When the phase shifts are chosen uniformly and the root sequences and weight sequences are generated by Zadoff-Chu (ZC) sequences [26], the interference among different UTs can be negligible as long as the length of weight sequences is large enough. Meanwhile, with GPSPs, the channel estimation can be implemented efficiently with much low computational complexity.

The main contributions of this work are summarized as follows:

- With 2D beam based channel model, we formulate a signal model for channel acquisition under the comb-type pilot structure and derive an equivalent representation with reduced-size matrices, thereby allowing a decrease in the computational complexity of channel estimation.
- We propose the GPSPs and reveal the constant modulus property for GPSPs and their discrete Fourier transform. Besides, we prove that the DFT of GPSPs has optimal autocorrelation property and beneficial crosscorrelation property. With the properties of GPSPs, the inter-user interference with GPSPs can be ignored when the length of weight sequences is large enough. Therefore, GPSPs offer a new choice for channel acquisition that does not rely on the statistical CSI.
- Since the length of weight sequences cannot be too long in practical systems, when spatial statistical CSI, which is easier to acquire, is available, we present a pilot scheduling algorithm to reduce the inter-user interference. Besides, when the SF statistical CSI is available, the conditions for optimal channel estimation are derived.
- Utilizing the properties of GPSPs, we propose an efficient implementation for GAMP based channel estimation, which can be generalized to a class of estimation methods. The proposed pilot provides a trade-off between performance and computational complexity. The performance of the proposed GPSPs is evaluated under a typical propagation scenario, and compared with ZC sequences and APSPs by simulation results.

The rest of this paper is organized as follows. In Section II, we present a signal model for channel acquisition within the comb-type pilot structure. In Section III, the GPSPs with relevant properties are proposed, and the inter-user interference resulted from GPSPs is analyzed. In Section IV, we investigate the channel estimation with GPSPs and present an efficient implementation by leveraging the properties of GPSPs. Simulation results are provided in Section V. The conclusion is drawn in Section VI.

Notations: The following notations are adopted in this paper. The uppercase (lowercase) boldface letters denote matrices (column vectors). $\bar{\jmath} = \sqrt{-1}$ is the imaginary unit. The notation $\triangleq$ is used for definitions. We use $\lceil\cdot\rceil$ ($\lfloor\cdot\rfloor$) to denote the rounding up (down). The notation $\langle\cdot\rangle_N$ is modulo $N$. The superscripts $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, $(\cdot)^{\mathrm{H}}$ denote the conjugate, transpose, conjugate-transpose, respectively. The operations $\odot$ and $\otimes$ denote the Hadamard product and Kronecker product, respectively. We use $[\mathbf{x}]_n$, $[\mathbf{X}]_{i,j}$, $[\mathbf{X}]_{i,:}$ and $[\mathbf{X}]_{i:j,m:n}$ to denote the $n$-th element of the vector $\mathbf{x}$, the $(i,j)$-th element, the $i$-th column and rows $i$ to $j$ and columns $m$ to $n$ of the matrix $\mathbf{X}$, respectively, with element indices starting with 0. $\mathrm{tr}(\mathbf{A})$ denotes the trace of $\mathbf{A}$. $\|\cdot\|_F$ denotes the Frobenius norm. $\mathrm{diag}(\mathbf{x})$ denotes the diagonal matrix with $\mathbf{x}$ along its diagonal. $\mathrm{Bdiag}(\mathbf{A}_0, \mathbf{A}_1, \ldots)$ denotes a block diagonal matrix with the elements $\mathbf{A}_i$ located along the main diagonal. $\mathbb{E}\{\cdot\}$ denotes the expectation operation and $\mathrm{vec}(\cdot)$ denotes the vectorization operation. $\mathbf{I}_N$ denotes the identity matrix of $N$ dimension while $\mathbf{I}_{a,b}$ denotes $\max(a,b)$-dimensional matrix with $[\mathbf{I}_{a,b}]_{n,n} = 1, n = 0, 1, \ldots, \min(a,b) - 1$ and the other elements being 0. $\mathbf{1}_N$ denotes the $N \times 1$ vector with all elements being 1. $\mathbf{F}_N$ denotes the $N$-dimensional DFT matrix. $\mathrm{circ}(\mathbf{x})$ is the circulant matrix whose first column is $\mathbf{x}$. We further define the cyclic shift matrix

$$\mathbf{\Pi}_N^a \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{I}_{N-\langle a\rangle_N} \\ \mathbf{I}_{\langle a\rangle_N} & \mathbf{0}. \end{bmatrix}.$$

## II. System Model

In this section, we present the system configuration of the massive MIMO-OFDM communication and the 2D beam based channel model. With 2D beam based channel model, we formulate a signal model for channel acquisition under comb-type pilot structure.

### A. System Configuration

Consider a massive MIMO-OFDM system operating in time division duplexing (TDD) mode, where the BS is equipped with a uniform linear array consisting of $M$ antennas, serving $U$ single antenna UTs. The spacing between adjacent antennas is denoted by $d = \frac{\lambda_c}{2}$, where $\lambda_c$ is the wavelength corresponding to the carrier frequency $f_c$.

In a single OFDM symbol, the number of subcarriers is $N_{\mathrm{sc}}$, the length of cyclic prefix (CP) is $N_{\mathrm{cp}}$, the subcarrier interval length is $\Delta f$, the sample interval length is $T_s = \frac{1}{N_{\mathrm{sc}}\Delta f}$. We assume that there are $N_v$ valid subcarriers with index starting from $k_0$ used for pilot and data transmission. Due

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to the reciprocity in TDD systems, we focus on the channel acquisition in uplink (UL) transmission.

### B. 2D Beam Based Channel Model

During the UL transmission, the received continuous-time baseband signal of the $i$-th antenna at BS can be expressed as [19], [27], [28]

$$y_i = \sum_{u=0}^{U-1} \int_{-\infty}^{\infty} h_{u,i}(t,\tau) x_u(t-\tau) d\tau + z_i(t), \tag{1}$$

where $x_u$ is the OFDM signal sent by UT $u$ and $z_i$ denotes the additive white Gaussian noise at the $i$-th antenna, and $h_{u,i}(t,\tau)$ is the time-variant channel impulse response between UT $u$ and the $i$-th antenna at the BS side. For convenience, we assume that the channel does not change over an OFDM symbol, and then it can be expressed as

$$h_{u,i}(\tau) = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{-\bar{\jmath}2\pi f_c \Delta\tau i \Omega_{u,p}} \delta(\tau - \tau_{u,p}), \tag{2}$$

where $P_u$ denotes the number of paths between the UT $u$ and the BS, $\gamma_{u,p}$ and $\tau_{u,p}$ are complex-valued gain and propagation delay at the first antenna of the $p$-th path between the UT $u$ and the BS, respectively, $\Delta\tau = d/c$ is the delay between adjacent antennas with $c$ being the speed of light, $\Omega_{u,p} = \sin\theta_{u,p}^{\mathrm{az}} \cos\theta_{u,p}^{\mathrm{el}}$ is the directional cosine for the $p$-th path between UT $u$ and the BS, with $\theta_{u,p}^{\mathrm{az}}$ and $\theta_{u,p}^{\mathrm{el}}$ representing the azimuth and the elevation angles of arrival (AoA), respectively. Then, the channel frequency response at the $k$-th subcarrier between the UT $u$ and the $i$-th antenna at the BS can be expressed as

$$\begin{aligned} h_{u,i,k}^{\mathrm{SF}} &= \int h_{u,i}(\tau) e^{-\bar{\jmath}2\pi k \Delta f \tau} d\tau \\ &= \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{-\bar{\jmath}2\pi k \Delta f \tau_{u,p}} e^{-\bar{\jmath}2\pi f_c \Delta\tau i \Omega_{u,p}}. \end{aligned} \tag{3}$$

Let $\vartheta_{u,p} \triangleq f_c \Delta\tau \Omega_{u,p} = \frac{1}{2}\Omega_{u,p}$ denote the scaled directional cosine, the steering vectors in space and frequency domain can be expressed as

$$\mathbf{v}_{\mathrm{s}}(\vartheta) \triangleq \left[1, e^{-\bar{\jmath}2\pi\vartheta}, \ldots, e^{-\bar{\jmath}2\pi(M-1)\vartheta}\right]^{\mathrm{T}}, \tag{4}$$

$$\mathbf{v}_{\mathrm{f}}(\tau) \triangleq \left[e^{-\bar{\jmath}2\pi k_0 \Delta f \tau}, e^{-\bar{\jmath}2\pi k_1 \Delta f \tau}, \ldots, e^{-\bar{\jmath}2\pi k_{N_{\mathrm{v}}-1} \Delta f \tau}\right]^{\mathrm{T}}, \tag{5}$$

which point towards scaled directional cosine $\vartheta$ and delay $\tau$, respectively. By means of these steering vectors, the channel between the UT $u$ and the BS in the SF domain is

$$\mathbf{H}_u^{\mathrm{SF}} \triangleq \sum_{p=0}^{P_u-1} \gamma_{u,p} \mathbf{v}_{\mathrm{s}}(\vartheta_{u,p}) \mathbf{v}_{\mathrm{f}}^{\mathrm{T}}(\tau_{u,p}), \tag{6}$$

where $[\mathbf{H}_u^{\mathrm{SF}}]_{i,k} = h_{u,i,k}^{\mathrm{SF}}$.

The ranges of parameters $\vartheta_{u,p}$ and $\tau_{u,p}$ are $\mathcal{R}_{\mathrm{an}} = \left[-\frac{1}{2}, \frac{1}{2}\right)$ and $\mathcal{R}_{\mathrm{de}} = \left[0, \frac{N_{\mathrm{cp}}}{N_{\mathrm{sc}}\Delta f}\right)$, which can be uniformly divided into $N_{\mathrm{an}}$ and $N_{\mathrm{de}}$ intervals, respectively, as follows:

$$\mathcal{R}_{\mathrm{an}} = \bigcup_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \Lambda_{n_{\mathrm{an}}}, \Lambda_{n_{\mathrm{an}}} = \left[\frac{n_{\mathrm{an}}}{N_{\mathrm{an}}} - \frac{1}{2}, \frac{n_{\mathrm{an}}+1}{N_{\mathrm{an}}} - \frac{1}{2}\right), \tag{7}$$

$$\mathcal{R}_{\mathrm{de}} = \bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \Lambda_{n_{\mathrm{de}}}, \Lambda_{n_{\mathrm{de}}} = \left[\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}} N_{\mathrm{v}} \Delta f}, \frac{N_\tau (n_{\mathrm{de}}+1)}{N_{\mathrm{de}} N_{\mathrm{v}} \Delta f}\right), \tag{8}$$

where $N_\tau = N_{\mathrm{v}} N_{\mathrm{cp}}/N_{\mathrm{sc}}$, $N_{\mathrm{an}} = F_{\mathrm{an}} M$ and $N_{\mathrm{de}} = F_{\mathrm{de}} N_\tau$ with $F_{\mathrm{an}}$ and $F_{\mathrm{de}}$ representing fine factors to flexibly adjust the density of oversampling. Let $\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}}} = \Lambda_{n_{\mathrm{an}}} \times \Lambda_{n_{\mathrm{de}}}$, where the operation $\times$ represents the Cartesian product of two sets. Denote $\Gamma_u \triangleq \{(\vartheta_{u,0}, \tau_{u,0}), (\vartheta_{u,1}, \tau_{u,1}), \ldots, (\vartheta_{u,P_u-1}, \tau_{u,P_u-1})\}$ as the set of parameter pairs of the UT $u$. Then, scaled directional cosine $\vartheta_{u,p}$ and delay $\tau_{u,p}$ must locate in one sampled interval $\Lambda_{n_{\mathrm{an}}}$ and $\Lambda_{n_{\mathrm{de}}}$, respectively. We approximate the true parameters located in a sampled interval with the left-hand side endpoint. Then the SF domain channel can be approximated as

$$\mathbf{H}_u^{\mathrm{SF}} \approx \mathbf{V}_{\mathrm{s}} \mathbf{H}_u \mathbf{V}_{\mathrm{f}}^{\mathrm{T}}, \tag{9}$$

where $\mathbf{H}_u \in \mathbb{C}^{N_{\mathrm{an}} \times N_{\mathrm{de}}}$ is the 2D beam domain channel matrix satisfying $[\mathbf{H}_u]_{n_{\mathrm{an}},n_{\mathrm{de}}} \triangleq \sum_{\Gamma_u \cap \Lambda_{n_{\mathrm{an}},n_{\mathrm{de}}}} \gamma_{u,p}$. $\mathbf{V}_{\mathrm{s}} \triangleq [\mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(0), \ldots, \mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(N_{\mathrm{an}}-1)]$ and $\mathbf{V}_{\mathrm{f}} \triangleq [\mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(0), \ldots, \mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(N_{\mathrm{de}}-1)]$ are the spatial beam matrix and the frequency beam matrix, respectively, where $\mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(n_{\mathrm{an}}) \triangleq \mathbf{v}_{\mathrm{s}}\left(\frac{n_{\mathrm{an}}}{N_{\mathrm{an}}} - \frac{1}{2}\right)$ is the sampled steering vector for the spatial beam and $\mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(n_{\mathrm{de}}) \triangleq \mathbf{v}_{\mathrm{f}}\left(\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}} N_{\mathrm{v}} \Delta f}\right)$ is the sampled steering vector for the frequency beam. The beam matrices can be rewritten as

$$\mathbf{V}_{\mathrm{s}} = \mathbf{I}_{M,N_{\mathrm{an}}} \mathbf{F}_{N_{\mathrm{an}}} \mathbf{\Pi}_{N_{\mathrm{an}}}^{-N_{\mathrm{an}}/2}, \tag{10}$$

$$\mathbf{V}_{\mathrm{f}} = \mathbf{I}_{N_{\mathrm{v}},N_{\mathrm{de}}^{\mathrm{v}}} \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{v}}} \mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{v}}}^{k_0} \mathbf{I}_{N_{\mathrm{de}}^{\mathrm{v}},N_{\mathrm{de}}}, \tag{11}$$

where $\mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{v}}}^{k_0} = \mathrm{diag}\left(\left[1, e^{\bar{\jmath}2\pi \frac{k_0}{N_{\mathrm{de}}^{\mathrm{v}}}}, \ldots, e^{\bar{\jmath}2\pi \frac{k_0 (N_{\mathrm{de}}^{\mathrm{v}}-1)}{N_{\mathrm{de}}^{\mathrm{v}}}}\right]\right)$ and $N_{\mathrm{de}}^{\mathrm{v}} = F_{\mathrm{de}} N_{\mathrm{v}}$.

Each element in $\mathbf{H}_u$ corresponds to a physical channel in the 2D beam domain. Typically, most of the elements are close to zero because of the limited number of local scatters [29]. Thus, the 2D beam domain channel matrix is sparse both in the angle dimension and the delay dimension, which greatly reduces the number of elements to be estimated and allows for channel estimation through sparse signal recovery. The true directional cosines and delays of paths generally do not lie exactly on discrete grid points, which leads to grid mismatch when transforming the channel into the 2D beam domain. As a consequence, beam domain elements originating from the same physical path are correlated. However, the 2D beam based channel representation can approach the physical channel with increased fine factors, and the beam domain elements tend to be independent in this case. Therefore, we assume that the elements in $\mathbf{H}_u$ follow the independent circular symmetric complex Gaussian distributions with zero mean and different variances. Previous works show that when the fine factors satisfy $F_{\mathrm{an}} \geq 2$ and $F_{\mathrm{de}} \geq 2$, the performance degradation caused by grid mismatch is significantly alleviated, and the independence approximation provides a satisfactory performance and tractable representation [19], [20]. Thus the 2D beam domain channel covariance matrix $\mathbf{R}_u \triangleq \mathbb{E}\{\mathrm{vec}(\mathbf{H}_u)\mathrm{vec}^{\mathrm{H}}(\mathbf{H}_u)\}$ is a diagonal matrix. Therefore,

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Fig. 1. Illustration of comb pilot structure.

the statistical CSI can be expressed by 2D beam domain channel power matrix $\mathbf{D}_u \triangleq \mathbb{E}\{\mathbf{H}_u \odot (\mathbf{H}_u)^*\}$, which contains all diagonal elements in $\mathbf{R}_u$. It should be noted that the 2D beam domain channel power matrix evolves on a much slower timescale than instantaneous CSI. Existing studies have shown that the beam domain power matrices exhibit strong temporal correlation over relatively long durations [30]. In addition, low-complexity approaches for acquiring beam domain statistical CSI have been proposed [21]. Thus, we assume that $\mathbf{D}_u$ of all users are known at the BS in the rest of the paper.

### C. Signal Model for Channel Acquisition

The comb-type pilot structure [31] over the frequency dimension is widely used in 5G wireless communication for frequency multiplexing [32]. Using all subcarriers can be treated as a special case of comb-type pilot structure, and the comb-type pilot is compatible with any frame structure used in TDD uplink channel estimation. Therefore, the comb-type structure is considered in this paper.

For comb-type pilot structure as shown in Fig. 1, valid subcarriers are divided into $N_{\mathrm{c}}$ evenly spaced subcarrier groups, which is referred to as the combs. Each of combs has $N_{\mathrm{v}}^{\mathrm{p}} = N_{\mathrm{v}}/N_{\mathrm{c}}$ subcarriers to transmit pilots of $U_{\mathrm{c}} = U/N_{\mathrm{c}}$ UTs. For simplicity, $N_{\mathrm{v}}$ and $U$ are assumed divisible by $N_{\mathrm{c}}$. Considering UTs using the $n_{\mathrm{c}}$-th comb, let $\mathbf{x}_u$ denote the pilot sequences sent by UT $u$ with length of $N_{\mathrm{v}}^{\mathrm{p}}$ over subcarriers. The received pilot signal at BS can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \sum_{u=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}}\mathbf{H}_u(\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}})^{\mathrm{T}}\mathbf{X}_u + \mathbf{Z}_{n_{\mathrm{c}}} \in \mathbb{C}^{M\times N_{\mathrm{v}}^{\mathrm{p}}}, \tag{12}$$

where $\mathbf{X}_u = \mathrm{diag}(\mathbf{x}_u)$, $\mathbf{Z}_{n_{\mathrm{c}}}$ denotes the circularly symmetric complex Gaussian noise with distribution $\mathcal{CN}(\mathrm{vec}(\mathbf{Z}_{n_{\mathrm{c}}}); 0, \sigma_z^2\mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}})$, $\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}} = \mathbf{S}_{N_{\mathrm{v}}}^{N_{\mathrm{c}},n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f}} \in \mathbb{C}^{N_{\mathrm{v}}^{\mathrm{p}}\times N_{\mathrm{de}}}$ is the frequency beam matrix for the $n_{\mathrm{c}}$-th comb, and $\mathbf{S}_N^{a,b} \in \mathbb{C}^{\frac{N}{a}\times N}$ is the selection matrix satisfying $\mathbf{S}_N^{a,b}\mathbf{A} = [\mathbf{A}_{b,:}^{\mathrm{T}}, \mathbf{A}_{a+b,:}^{\mathrm{T}}, \ldots, \mathbf{A}_{N-a+b,:}^{\mathrm{T}}]^{\mathrm{T}}$, for $\mathbf{A}$ with $N$ rows. The matrix $\mathbf{S}_N^{a,b}$ can be constructed by setting $[\mathbf{S}_N^{a,b}]_{i,ai+b} = 1, i = 0, 1, \ldots, N/a-1$ and the other elements to 0. Assuming $N_{\mathrm{de}} \le N_{\mathrm{de}}^{\mathrm{p}}$ with $N_{\mathrm{de}}^{\mathrm{p}} = F_{\mathrm{de}}N_{\mathrm{v}}^{\mathrm{p}}$, and then, with (11), $\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}}$ can be rewritten as

$$\begin{aligned}\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}} &= \mathbf{S}_{N_{\mathrm{v}}}^{N_{\mathrm{c}},n_u}\mathbf{I}_{N_{\mathrm{v}},N_{\mathrm{de}}^{\mathrm{v}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{v}}}\mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{v}}}^{k_0}\mathbf{I}_{N_{\mathrm{de}}^{\mathrm{v}},N_{\mathrm{de}}} \\ &= \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{S}_{N_{\mathrm{de}}^{\mathrm{v}}}^{N_{\mathrm{c}},n_u}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{v}}}\mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{v}}}^{k_0}\mathbf{I}_{N_{\mathrm{de}}^{\mathrm{v}},N_{\mathrm{de}}} \\ &= \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}^{N_{\mathrm{c}}}\mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{v}}}^{k_{n_{\mathrm{c}}}}\mathbf{I}_{N_{\mathrm{de}}^{\mathrm{v}},N_{\mathrm{de}}} \\ &= \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}\bar{\mathbf{\Phi}}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}}\mathbf{I}_{N_{\mathrm{de}}^{\mathrm{p}},N_{\mathrm{de}}},\end{aligned} \tag{13}$$

where $\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}^{N_{\mathrm{c}}} \triangleq [\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}, \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}, \ldots, \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}] \in \mathbb{C}^{N_{\mathrm{de}}^{\mathrm{p}}\times N_{\mathrm{de}}^{\mathrm{v}}}$, and $\bar{\mathbf{\Phi}}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}} = \mathrm{diag}\left(\left[1, e^{\bar{\jmath}2\pi\frac{k_{n_{\mathrm{c}}}}{N_{\mathrm{de}}^{\mathrm{v}}}}, \ldots, e^{\bar{\jmath}2\pi\frac{k_{n_{\mathrm{c}}}(N_{\mathrm{de}}^{\mathrm{p}}-1)}{N_{\mathrm{de}}^{\mathrm{v}}}}\right]\right)$. Therefore, 2D beam domain channel matrix and frequency beam matrix can be expressed in the following way

$$\mathbf{H}_u^{n_{\mathrm{c}}} \triangleq \mathbf{H}_u\mathbf{I}_{N_{\mathrm{de}},N_{\mathrm{de}}^{\mathrm{p}}}\bar{\mathbf{\Phi}}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}} \in \mathbb{C}^{N_{\mathrm{an}}\times N_{\mathrm{de}}^{\mathrm{p}}}, \tag{14}$$

$$\mathbf{V}_{\mathrm{f,p}} \triangleq \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}} \in \mathbb{C}^{N_{\mathrm{v}}^{\mathrm{p}}\times N_{\mathrm{de}}^{\mathrm{p}}}, \tag{15}$$

where the relation $\mathbf{H}_u(\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}})^{\mathrm{T}} = \mathbf{H}_u^{n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f,p}}^{\mathrm{T}}$ holds when $N_{\mathrm{de}} \le N_{\mathrm{de}}^{\mathrm{p}}$. Then with (14) and (15), the received pilot signal can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \sum_{u=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}}\mathbf{H}_u^{n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f,p}}^{\mathrm{T}}\mathbf{X}_u + \mathbf{Z}_{n_{\mathrm{c}}}. \tag{16}$$

Additionally, since $\mathbf{R}_u$ and $\bar{\mathbf{\Phi}}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}}$ are both diagonal, the covariance matrix of the equivalent 2D beam domain channel $\mathbf{H}_u^{n_{\mathrm{c}}}$ is the diagonal matrix $\mathbf{R}_u^{n_{\mathrm{c}}} \triangleq \mathbb{E}\{\mathrm{vec}(\mathbf{H}_u^{n_{\mathrm{c}}})\mathrm{vec}^{\mathrm{H}}(\mathbf{H}_u^{n_{\mathrm{c}}})\}$, which is diagonal and can be expressed by channel power matrix as $\mathbf{D}_u^{n_{\mathrm{c}}} \triangleq \mathbb{E}\{\mathbf{H}_u^{n_{\mathrm{c}}} \odot (\mathbf{H}_u^{n_{\mathrm{c}}})^*\}$. Then, we have $[\mathbf{D}_u^{n_{\mathrm{c}}}]_{:,0:N_{\mathrm{de}}-1} = \mathbf{D}_u$ and other elements in $\mathbf{D}_u^{n_{\mathrm{c}}}$ are 0.

Note that the received pilot signal can be rewritten as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \mathbf{V}_{\mathrm{s}}\mathbf{H}^{n_{\mathrm{c}}}(\mathbf{X}^{n_{\mathrm{c}}})^{\mathrm{T}} + \mathbf{Z}_{n_{\mathrm{c}}}, \tag{17}$$

where $\mathbf{X}^{n_{\mathrm{c}}} = [\mathbf{X}_0\mathbf{V}_{\mathrm{f,p}}, \mathbf{X}_1\mathbf{V}_{\mathrm{f,p}}, \ldots, \mathbf{X}_{U_{\mathrm{c}}-1}\mathbf{V}_{\mathrm{f,p}}]$ and $\mathbf{H}^{n_{\mathrm{c}}} = [\mathbf{H}_0^{n_{\mathrm{c}}}, \mathbf{H}_1^{n_{\mathrm{c}}}, \ldots, \mathbf{H}_{U_{\mathrm{c}}-1}^{n_{\mathrm{c}}}]$. With vectorization, the received pilot signal can be rewritten as

$$\mathbf{y}_{n_{\mathrm{c}}} = \mathbf{A}_{n_{\mathrm{c}}}\mathbf{h}_{n_{\mathrm{c}}} + \mathbf{z}_{n_{\mathrm{c}}}, \tag{18}$$

where $\mathbf{A}_{n_{\mathrm{c}}} \triangleq (\mathbf{X}^{n_{\mathrm{c}}})^{\mathrm{T}} \otimes \mathbf{V}_{\mathrm{s}} \in \mathbb{C}^{MN_{\mathrm{v}}^{\mathrm{p}}\times N_{\mathrm{an}}N_{\mathrm{de}}^{\mathrm{p}}}$, $\mathbf{y}_{n_{\mathrm{c}}}$, $\mathbf{h}_{n_{\mathrm{c}}}$ and $\mathbf{z}_{n_{\mathrm{c}}}$ are vectorizations of $\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}}$, $\mathbf{H}^{n_{\mathrm{c}}}$ and $\mathbf{Z}_{n_{\mathrm{c}}}$, respectively. Then $\mathbf{R}_{n_{\mathrm{c}}} = \mathbb{E}\{\mathbf{h}_{n_{\mathrm{c}}}\mathbf{h}_{n_{\mathrm{c}}}^{\mathrm{H}}\}$ is the covariance matrix of $\mathbf{h}_{n_{\mathrm{c}}}$, which is a diagonal matrix with $\mathbf{r}_{n_{\mathrm{c}}} = \mathrm{vec}(\mathbf{D}_u^{n_{\mathrm{c}}})$ located along its diagonal.

## III. Generalized Phase Shift Pilots

In this section, we propose generalized phase shift pilots (GPSPs) and reveal their relevant properties. Subsequently, we analyze the inter-user interference caused by GPSPs and present conditions for negligible interference when statistical CSI is unavailable or only spatial statistical CSI is available, respectively. Then conditions for optimal channel estimation with GPSPs are derived when SF statistical CSI is available.

### A. GPSP

Before presenting GPSPs, PSPs, which are a class of widely used pilots, should be introduced first. A phase shift pilot with length $N$ can be expressed as

$$\mathbf{x}_u^{\mathrm{p}} \triangleq \mathbf{x}_{\mathrm{r}} \odot \mathbf{f}_\phi, \tag{19}$$

where $\mathbf{x}_{\mathrm{r}}$ is the root sequence with length $N$ shared by all users, $\mathbf{f}_\phi = [\mathbf{F}_N]_{:,\phi}$ and $\phi$ is the phase shift factor. Utilizing the fact that $\mathbf{F}_N\mathbf{D}_\phi = \mathbf{\Pi}_N^\phi\mathbf{F}_N$, where $\mathbf{D}_\phi = \mathrm{diag}(\mathbf{f}_\phi)$, PSPs allow channel matrices to delicately shift along the delay

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dimension to achieve the optimal estimation performance [14], [33]. Then, the generalized phase shift pilots are defined as

$$\mathbf{x}_u \triangleq \mathbf{X}_{\mathrm{r}}\mathbf{\Phi}_{\mathrm{ps}}\mathbf{w}_u, \tag{20}$$

where $\mathbf{X}_{\mathrm{r}} = \mathrm{diag}(\mathbf{x}_{\mathrm{r}})$, $\mathbf{\Phi}_{\mathrm{ps}} = [\mathbf{f}_{\phi_0}, \mathbf{f}_{\phi_1} \ldots, \mathbf{f}_{\phi_{m-1}}]$, $\mathbf{w}_u$ is the $m \times 1$ vector composed of the weight sequence for the $u$-th UT, $m$ is the number of used phase shift factors and $\phi_k$ is the $k$-th chosen phase shift factor. Thus, GPSPs are linear combinations of multiple PSPs with different weights, and can be rewritten as

$$\mathbf{x}_u = \mathbf{x}_{\mathrm{r}} \odot \mathbf{x}_u^{\mathrm{ps}}, \tag{21}$$

where $\mathbf{x}_u^{\mathrm{ps}} = \sum_{k=0}^{m-1}[\mathbf{w}_u]_k \mathbf{f}_{\phi_k}$. Note that GPSPs are consistent with the definition of PSPs in (19) while the phase shift sequences $\mathbf{x}_u^{\mathrm{ps}}$ can be generated from an $m$-dimensional subspace formed by the columns of $\mathbf{\Phi}_{\mathrm{ps}}$. Therefore, GPSPs universally generalize the class of PSPs. Specially, when $\mathbf{\Phi}_{\mathrm{ps}} = \mathbf{F}_N$ and $\mathbf{w}_u$ is a vector with one element being 1 and the others being 0, the GPSPs are reduced to APSPs.

Considering the distinct properties of ZC sequences [34], [35], we use the Zadoff-Chu (ZC) sequence [26] for the root sequence $\mathbf{x}_{\mathrm{r}}$, i.e.

$$[\mathbf{x}_N^r]_n \triangleq e^{-\bar{\jmath}\pi \frac{rn(n+\langle N\rangle_2)}{N}}, \ n = 0, 1, 2, \ldots, N-1, \tag{22}$$

where the root $r$ is an integer relatively prime to $N$, and the weight sequence $\mathbf{w}_u$ is set to a ZC sequence with length of $m$ and root $q$. Then, to leverage the strengths of ZC sequences, the phase shift factors are chosen uniformly, and for simplicity, $\phi_0$ is set to 0. In this case, $s = N/m$ is referred to as the spacing length of the fixed phase shift factors. Denote $\tilde{\mathbf{x}}_u = \frac{1}{\sqrt{N}}\mathbf{F}_N^{\mathrm{H}}\mathbf{x}_u$, $\tilde{\mathbf{x}}_{\mathrm{r}} = \frac{1}{\sqrt{N}}\mathbf{F}_N^{\mathrm{H}}\mathbf{x}_{\mathrm{r}}$, and when $\mathbf{x}_{\mathrm{r}}$ is a ZC sequence, $\tilde{\mathbf{x}}_{\mathrm{r}}$ is also a ZC sequence with a new root $\tilde{r}$ [34], [35]. Then we have the following theorem.

*Theorem 1:* When the root sequence $\mathbf{x}_{\mathrm{r}}$ and the weight sequence $\mathbf{w}_u$ are both ZC sequences and $q + s\tilde{r}$ is relatively prime to $m$, $\mathbf{x}_u$ and its IDFT both have constant modulus with a modulus of $\sqrt{m}$.

*Proof:* See in Appendix A. ■

From Theorem 1, GPSPs have the same property of ZC sequences that both themselves and their IDFT have constant modulus, which is one of the prime reasons for the wide utilization of ZC sequences. Specifically, in an OFDM system, when GPSPs are applied in the frequency domain, the low peak to average power ratio (PAPR) is guaranteed directly in the time domain, which allows higher average transmit power and then facilitates the channel estimation [36]. Therefore, it is reasonable to set the root sequence and weight sequence to ZC sequences and choose phase shifts uniformly. Besides, $q + s\tilde{r}$ is assumed relatively prime to $m$ in subsequent discussions. Then, similar to ZC sequences, the IDFT of GPSPs has useful correlation properties, as well. The autocorrelation function can be defined as

$$R_a[p] = \tilde{\mathbf{x}}_u^{\mathrm{H}}\mathbf{\Pi}_N^p\tilde{\mathbf{x}}_u, \tag{23}$$

and the crosscorrelation function can be defined as

$$R_c^{q,q'}[p] = \tilde{\mathbf{x}}_{u'}^{\mathrm{H}}\mathbf{\Pi}_N^p\tilde{\mathbf{x}}_u, \tag{24}$$

where $q$, $q'$ are roots of $\mathbf{w}_u$, $\mathbf{w}_{u'}$, respectively, and $q \neq q'$. Then for the correlation properties of the IDFT of GPSPs, we have the following theorem.

*Theorem 2:* The autocorrelation function of the IDFT of GPSPs satisfies

$$R_a[p] = mN\delta[p], \tag{25}$$

and the modulus of crosscorrelation function satisfies

$$|R_c^{q,q'}[p]| = \sum_{k=0}^{m-1}\sqrt{m}N\delta[p - ks]. \tag{26}$$

*Proof:* See in Appendix B. ■

From Theorem 2, the IDFT of GPSPs has the optimal autocorrelation property, and the modulus of elements in crosscorrelation function does not exceed $1/\sqrt{m}$ of the peak of autocorrelation function. Therefore, when $m$ is large enough, the correlation between different GPSPs can be negligible, which enables GPSPs to behave similarly to orthogonal pilots and therefore benefits the channel estimation.

Then, according to the above theorems, GPSPs can be specified as below.

$$\mathbf{x}_u = \mathbf{x}_{N,m}^{r,q_u} = \mathbf{x}_N^r \odot \mathbf{x}_{m,N}^{q_u}, \tag{27}$$

where $[\mathbf{x}_{m,N}^{q_u}]_n \triangleq [\mathbf{x}_m^{q_u}]_{\langle n\rangle_m}, n = 0, 1, 2, \ldots, N-1$, with $N$ specified as $N_{\mathrm{v}}^{\mathrm{p}}$, and $q_u$ is the root for ZC sequence $\mathbf{x}_m^{q_u}$. In this case, the parameter $s$ represents the number of periodic extension of $\mathbf{x}_m^{q_u}$ in the $\mathbf{x}_{m,N}^{q_u}$.

**Remark:** In the proposed GPSP framework, the weight sequence $\mathbf{w}_u$ is not restricted to a specific realization in principle. However, the constant-modulus and correlation properties of Zadoff–Chu (ZC) sequences are can enable beneficial theoretical properties of GPSPs used in the analysis. When randomly generated unit modulus sequences are used as weights, these properties are generally not satisfied, and the resulting pilots deviate from the ideal GPSP structure, leading to performance degradation. For this reason, ZC-based weight sequences are adopted throughout this work.

### B. Inter-User Interference Analysis

Considering the channel estimation in a single cell, $\mathbf{x}_{N_{\mathrm{v}}^{\mathrm{p}}}^r$ is just the same for all users, while $\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u}$ changes for different users in the same comb and is reused for users in different combs. With the GPSPs, the received signal model can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \sum_{u=0}^{U_{\mathrm{c}}-1}\mathbf{V}_{\mathrm{s}}\mathbf{H}_u^{n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f,p}}^{\mathrm{T}}\mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u}\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r + \mathbf{Z}_{n_{\mathrm{c}}}, \tag{28}$$

where $\mathbf{X}_u$ in (12) is substituted by $\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r\mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u}$ with $\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r = \mathrm{diag}(\mathbf{x}_{N_{\mathrm{v}}^{\mathrm{p}}}^r)$ and $\mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u} = \mathrm{diag}(\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u})$. Then $q_u$ and $q_{u'}$ are used to distinguish weight sequences of different users in subsequent discussions.

We post-multiply the received signal by the conjugate transpose of the pilot matrix, which can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}}\mathbf{X}_u^{\mathrm{H}} = \sum_{u'=0}^{U_{\mathrm{c}}-1}\mathbf{V}_{\mathrm{s}}\mathbf{H}_{u'}^{n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f,p}}^{\mathrm{T}}\mathbf{X}_{u'}\mathbf{X}_u^{\mathrm{H}} + \widetilde{\mathbf{Z}}_{n_{\mathrm{c}}}, \tag{29}$$

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where $\widetilde{\mathbf{Z}}_{n_c} \triangleq \mathbf{Z}_{n_c}\mathbf{X}_u^{\mathrm{H}}$ is still the circularly symmetric complex Gaussian noise. Note that $\mathbf{X}_u\mathbf{V}_{\mathrm{f,p}}$ can be rewritten as

$$\mathbf{X}_u\mathbf{V}_{\mathrm{f,p}} = \mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{V}_{\mathrm{f,p}}\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}), \tag{30}$$

where $\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}} = \frac{1}{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}^{\mathrm{H}}_{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{x}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}$ and the detailed derivation is given in Appendix C. From (30), $\mathbf{X}_u$ can be split into $\mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}$ in frequency domain and $\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}})$ in delay domain, respectively. Substituting (30) into (29), we have

$$\begin{aligned}\mathbf{Y}^{\mathrm{SF}}_{n_c}\mathbf{X}_u^{\mathrm{H}} &= \sum_{u'=0}^{U_c-1}\mathbf{V}_{\mathrm{s}}\mathbf{H}^{n_c}_{u'}\mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}}(\mathbf{X}^{q_u}_{m,N_{\mathrm{v}}^{\mathrm{p}}})^{\mathrm{H}}\mathbf{X}^{q_{u'}}_{m,N_{\mathrm{v}}^{\mathrm{p}}} + \widetilde{\mathbf{Z}}_{n_c}\\ &= \sum_{u'=0}^{U_c-1}\mathbf{V}_{\mathrm{s}}\mathbf{H}^{n_c}_{u'}(\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}\mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}} + \widetilde{\mathbf{Z}}_{n_c},\end{aligned} \tag{31}$$

where $\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}) \triangleq (\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{H}}\mathrm{circ}(\widetilde{\mathbf{x}}^{q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}})$ as the product of two circulant matrices is also a circulant matrix and $[\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}]_n \triangleq (\widetilde{\mathbf{x}}^{q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}})^{\mathrm{H}}\mathbf{\Pi}^n_{N_{\mathrm{de}}^{\mathrm{p}}}\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}$. Then utilizing Theorem 2, (31) can be rewritten as

$$\mathbf{Y}^{\mathrm{SF}}_{n_c}\mathbf{X}_u^{\mathrm{H}} = \mathbf{V}_{\mathrm{s}}\left(\mathbf{H}^{n_c}_u + \sum_{u'\neq u}\mathbf{H}^{n_c}_{u'}(\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}\right)\mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}} + \widetilde{\mathbf{Z}}_{n_c}, \tag{32}$$

where the modulus of elements in $(\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}$ are no more than $1/\sqrt{m}$ when $q_{u'} \neq q_u$. From (32), we can define the inter-user interference of UT $u$ as

$$\mathbf{I}^{\mathrm{B}}_{n_c,u} = \sum_{u'\neq u}\mathbf{H}^{n_c}_{u'}(\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}. \tag{33}$$

Then we have the following theorem.

*Theorem 3:* When $m \to \infty$ and the number of propagation paths is limited, for the inter-user interference caused by GPSPs, we have

$$\lim_{m\to\infty}[\mathbf{I}^{\mathrm{B}}_{n_c,u}]_{n_{\mathrm{an}},n_{\mathrm{de}}} = 0, \tag{34}$$

for all $n_{\mathrm{an}}$ and $n_{\mathrm{de}}$.

*Proof:* See in Appendix D. ■

Theorem 3 suggests that when the length of weight sequences is sufficiently long, the inter-user interference can be negligible, thanks to the crosscorrelation properties of GPSPs. Although Theorem 3 characterizes the asymptotic decay trend on $m$, moderate values of $m$ also provide satisfactory interference suppression, as shown in the subsequent simulations. In this way, increasing $m$ is beneficial to suppress the inter-user interference since the modulus of interference item is no more than $1/\sqrt{m}$, which is monotonically decreasing with $m$. Meanwhile, Theorem 3 indicates the feasibility of the GPSPs in channel estimation without statistical CSI since it does not depend on any pilot scheduling algorithms. Besides, (33) indicates that inter-user interference increases as channel sparsity decreases. Therefore, in practical systems, the weight sequence length $m$ should be flexibly configured, and the number of served UTs should match the specific scenario to achieve satisfactory performance.

Moreover, since $m$ is not too large, when the statistical CSI is available, the statistical CSI can be used to decrease inter-user interference and improve channel estimation performance.

**Algorithm 1** Pilot Scheduling Algorithm With GPSPs

**Input:** The index set of UTs $\mathcal{U} = [0, 1, \ldots, U-1]$; the degree of overlap in angle dimension between UTs $\{\eta^{\mathrm{an}}_{p,u,u'}, u, u' \in \mathcal{U}\}$.

**Initialize:** $\forall u \in \mathcal{U}$, the assigned comb index $g_u = -1$;

1: **for** $p = 0 : N_{\mathrm{c}} - 1$ **do**
2: Select UT $u \in \{u|g_u = 0\}$ randomly, let $g_u = p$;
3: **for** $l = 1 : (U_{\mathrm{c}} - 1)$ **do**
4: Selecting UT $u \in \{u|g_u = 0\}$ minimizing $\sum_{u'}\eta^{\mathrm{an}}_{p,u,u'}, u' \in \{u|g_u = p\}$, let $g_u = p$;
5: **end for**
6: **end for**
7: **return** $\{g_u|u \in \mathcal{U}\}$.

As the AoA is relatively easier to obtain in practical systems [37], the case with only spatial statistical CSI is considered here. Define the degree of channel overlap between UT $u$ and $u'$ in the angle dimension as

$$\eta^{\mathrm{an}}_{n_c,u,u'} \triangleq \frac{(\mathbf{d}^{n_c}_{u'})^{\mathrm{H}}\mathbf{d}^{n_c}_u}{\|\mathbf{d}^{n_c}_{u'}\|\|\mathbf{d}^{n_c}_u\|}, \tag{35}$$

where $\mathbf{d}^{n_c}_u = \mathbf{D}^{n_c}_u\mathbf{1}_{N_{\mathrm{v}}^{\mathrm{p}}}$ is the spatial statistical CSI. Note that the angle dimension of $\mathbf{H}^{n_c}_u$ is sparse due to the small angle spread and the limited number of propagation paths. We define the indexes set $\mathbb{P}^{n_c}_{\mathrm{an},u} = \{n_{\mathrm{an}}|[\mathbf{H}^{n_c}_u]_{n_{\mathrm{an}},:} \neq \mathbf{0}\}$ containing all spatial beams with propagated signals of UT $u$. Then, selection matrix $\mathbf{S}^{n_c}_u$ is constructed by letting $[\mathbf{S}^{n_c}_u]_{i,i} = 1, i \in \mathbb{P}^{n_c}_{\mathrm{an},u}$ and the other elements be 0. Then we have the following theorem.

*Theorem 4:* When $\forall u \neq u', \eta^{\mathrm{an}}_{n_c,u,u'} = 0$, the 2D beam domain channel of $u$ can be distinguished from interference in the angle dimension as

$$\mathbf{H}^{n_c}_u = \mathbf{S}^{n_c}_u(\mathbf{H}^{n_c}_u + \mathbf{I}^{\mathrm{B}}_{n_c,u}). \tag{36}$$

*Proof:* Since $\forall u \neq u', \eta^{\mathrm{an}}_{n_c,u,u'} = 0$, for a fixed $u$ and for all $n_{\mathrm{an}} \in \mathbb{P}^{n_c}_{\mathrm{an},u}$, we have $\mathbf{H}^{n_c}_u = \mathbf{S}^{n_c}_u\mathbf{H}^{n_c}_u$ and $[\mathbf{H}^{n_c}_{u'}(\mathrm{circ}(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}]_{n_{\mathrm{an}},:} = \mathbf{0}^{\mathrm{T}}$ as $[\mathbf{H}^{n_c}_{u'}]_{n_{\mathrm{an}},:} = \mathbf{0}^{\mathrm{T}}$. Thus, we have $\mathbf{H}^{n_c}_u = \mathbf{S}^{n_c}_u\mathbf{H}^{n_c}_u$. ■

Theorem 4 suggests that when GPSPs are used in channels with no overlap in angle dimension, the inter-user interference can be totally removed by utilizing the spatial statistical CSI after getting the 2D beam domain channel, which is in agreement with the principle of spatial multiplexing.

Motivated by Theorem 4, we can utilize the degree of channel overlap to perform a pilot scheduling, which is summarized in Algorithm 1. The output $g_u$ is the assigned comb index of UT $u$. The algorithm allocates UTs into different combs to minimize the average degree of channel overlap in the angle dimension between UTs in the same comb with a greedy strategy. The computational complexity of Algorithm 1 is $\mathcal{O}(N_{\mathrm{an}}U^2 + U^2)$, where the term $\mathcal{O}(N_{\mathrm{an}}U^2)$ comes from calculating the degree of channel overlap in the angle domain, and the term $\mathcal{O}(U^2)$ comes from the UT grouping. Note that the complexity of the scheduling algorithm is significantly lower than that of channel estimation, and scheduling is performed at a much slower timescale than channel estimation. As a result, the overall system complexity is dominated by

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the channel estimation stage, and the scheduling overhead is negligible in comparison.

### C. Channel Estimation Performance Analysis

As shown in above, GPSPs can work well without statistical CSI or with spatial statistical CSI. Furthermore, they can also be applied to the case with SF statistical CSI. In this case, the optimal channel estimation performance with GPSPs can be analyzed based on MMSE channel estimation. Then, from (18), the MMSE estimate of $\mathbf{h}_{n_c}$ can be expressed as

$$\widehat{\mathbf{h}}_{n_c}^{\mathrm{MMSE}} = \mathbf{R}_{n_c}(\mathbf{A}_{n_c}^{\mathrm{H}}\mathbf{A}_{n_c}\mathbf{R}_{n_c} + \sigma_z^2\mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}})^{-1}\mathbf{A}_{n_c}^{\mathrm{H}}\mathbf{y}_{n_c}. \quad (37)$$

Then denote $\widehat{\mathbf{H}}_u^{n_c}$ as the estimated $\mathbf{H}_u^{n_c}$ extracted from $\widehat{\mathbf{h}}_{n_c}^{\mathrm{MMSE}}$. According to (14), the estimated 2D beam domain channel of UT $u$ can be obtained as

$$\widehat{\mathbf{H}}_u = \widehat{\mathbf{H}}_u^{n_c}(\bar{\mathbf{\Phi}}_{N_{\mathrm{v}}^{\mathrm{de}}}^{k_{n_c}})^{\mathrm{H}}\mathbf{I}_{N_{\mathrm{de}}^{\mathrm{p}},N_{\mathrm{de}}}. \quad (38)$$

Then according to (9), the estimated SF domain channel $\widehat{\mathbf{H}}_u^{\mathrm{SF}}$ can be recovered as

$$\widehat{\mathbf{H}}_u^{\mathrm{SF}} = \mathbf{V}_{\mathrm{s}}\widehat{\mathbf{H}}_u\mathbf{V}_{\mathrm{f}}^{\mathrm{T}}. \quad (39)$$

Let $\widehat{\mathbf{H}}_{n_c,u}^{\mathrm{SF}} = \widehat{\mathbf{H}}_u^{\mathrm{SF}}(\mathbf{S}_{N_{\mathrm{v}}}^{N_{\mathrm{c}},n_c})^{\mathrm{T}}$ and $\mathbf{H}_{n_c,u}^{\mathrm{SF}} = \mathbf{H}_u^{\mathrm{SF}}(\mathbf{S}_{N_{\mathrm{v}}}^{N_{\mathrm{c}},n_c})^{\mathrm{T}}$ denote the estimated and real-time channel of UT $u$ on the subcarriers of the $n_{\mathrm{c}}$-th comb in SF domain, respectively, and with vectorization, $\widehat{\mathbf{h}}_{n_c,u}^{\mathrm{SF}} = \mathrm{vec}(\widehat{\mathbf{H}}_{n_c,u}^{\mathrm{SF}})$ and $\mathbf{h}_{n_c,u}^{\mathrm{SF}} = \mathrm{vec}(\mathbf{H}_{n_c,u}^{\mathrm{SF}})$. As an important performance metric of channel estimation, the mean square error (MSE) of the estimated channel in the $n_{\mathrm{c}}$-th comb is

$$\begin{aligned}\epsilon &\triangleq \mathbb{E}\{\|\widehat{\mathbf{h}}_{n_c}^{\mathrm{SF}} - \mathbf{h}_{n_c}^{\mathrm{SF}}\|^2\} \\ &= \sum_{u=0}^{U_{\mathrm{c}}-1}\mathrm{tr}(\mathbf{R}_{n_c,u}^{\mathrm{SF}} - \mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\mathbf{C}_{n_c}^{-1}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}), \end{aligned} \quad (40)$$

where $\mathbf{h}_{n_c}^{\mathrm{SF}} = [(\mathbf{h}_{n_c,0}^{\mathrm{SF}})^{\mathrm{T}}, (\mathbf{h}_{n_c,1}^{\mathrm{SF}})^{\mathrm{T}}, \ldots, (\mathbf{h}_{n_c,U_{\mathrm{c}}-1}^{\mathrm{SF}})^{\mathrm{T}}]^{\mathrm{T}}$, $\widehat{\mathbf{h}}_{n_c}^{\mathrm{SF}} = [(\widehat{\mathbf{h}}_{n_c,0}^{\mathrm{SF}})^{\mathrm{T}}, (\widehat{\mathbf{h}}_{n_c,1}^{\mathrm{SF}})^{\mathrm{T}}, \ldots, (\widehat{\mathbf{h}}_{n_c,U_{\mathrm{c}}-1}^{\mathrm{SF}})^{\mathrm{T}}]^{\mathrm{T}}$, $\mathbf{R}_{n_c,u}^{\mathrm{SF}} \triangleq \mathbb{E}\{\mathbf{h}_{n_c,u}^{\mathrm{SF}}(\mathbf{h}_{n_c,u}^{\mathrm{SF}})^{\mathrm{H}}\}$, $\mathbf{C}_{n_c} \triangleq \sum_{u=0}^{U_{\mathrm{c}}-1}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}} + \sigma_z^2\mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}}$, and $\bar{\mathbf{X}}_u = \mathbf{X}_u \otimes \mathbf{I}_M$. Then we have the following theorem, which provides optimal conditions to minimize $\epsilon$.

*Theorem 5:* The minimum value of $\epsilon$ is given by

$$\epsilon_{\min} \approx \sum_{u=0}^{U_{\mathrm{c}}-1}\mathrm{tr}(\mathbf{R}_{n_c,u}^{\mathrm{SF}} - \mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}(\mathbf{R}_{n_c,u}^{\mathrm{SF}} + \sigma_z^2\mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}})^{-1}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}), \quad (41)$$

when either of the following conditions is satisfied:

1) $M$ is sufficiently large and for $\forall u \neq u'$ in the $n_{\mathrm{c}}$-th comb $\eta_{n_c,u,u'}^{\mathrm{an}} = 0$,
2) $m$ is sufficiently large.

*Proof:* See in Appendix E. ■

From Theorem 5, the minimum MSE $\epsilon_{\min}$ is the same as the sum of $U_{\mathrm{c}}$ respective MSEs of estimated channels in single-user channel estimation. It implies that the inter-user interference under GPSPs is removable when any of the conditions above is satisfied, and the channel estimation can achieve the optimal performance in the sense of the MSE. The Condition 1) of Theorem 5 also provides an insight that the performance of the GPSPs can be improved by an efficient pilot scheduling algorithm utilizing the spatial statistical CSI, which is consistent with Theorem 4. The Condition 2) implies that the inter-user interference caused by GPSPs is negligible when $m$ is sufficiently large, which is also in agreement with Theorem 3.

## IV. Channel Estimation and Efficient Implementation

In this section, we first present a GAMP based channel estimation method generalized to the case with SF statistical CSI, with only spatial statistical CSI, or without statistical CSI. Then, we propose an efficient implementation for channel estimation, which significantly reduces the computational complexity.

### A. GAMP Based Channel Estimation

For the channel estimation with the SF statistical CSI, the widely used GAMP algorithm, which can obtain MMSE estimates with only matrix multiplications and element-wise operations, is chosen as an example framework. Applying the GAMP algorithm for the channel estimation with (18), the mean $\hat{\mathbf{h}}_{n_c}^t$ and variance $\boldsymbol{\tau}_x^t$ of the posteriori estimate of $\mathbf{h}_{n_c}$ at the $t$-th iteration can be expressed by

$$\hat{\mathbf{h}}_{n_c}^t = \mathbf{r}_{n_c}^{t-1} \odot [\hat{\mathbf{h}}_{n_c}^{t-1} + \boldsymbol{\tau}_r^{t-1} \odot (\mathbf{A}_{n_c}^{\mathrm{H}}\mathbf{y}_{n_c}^{t-1})], \quad (42)$$

$$\boldsymbol{\tau}_x^t = \mathbf{r}_{n_c}^{t-1} \odot \boldsymbol{\tau}_r^{t-1}, \quad (43)$$

where $\mathbf{r}_{n_c}^{t-1} = \mathbf{r}_{n_c} \odot (\mathbf{r}_{n_c} + \boldsymbol{\tau}_r^{t-1})^{\circ -1}$, the superscript $t$ denotes the iteration index, $(\cdot)^{\circ -1}$ denotes the element-wise inversion, and $\mathbf{y}_{n_c}^{t-1}$ and $\boldsymbol{\tau}_r^{t-1}$ are computed as

$$\mathbf{y}_{n_c}^{t-1} = (\mathbf{y}_{n_c} - \boldsymbol{\mu}_p^{t-1}) \odot (\sigma_z^2\mathbf{1}_{M_{\mathrm{all}}^{\mathrm{p}}} + \boldsymbol{\tau}_p^{t-1})^{\circ -1}, \quad (44)$$

$$\boldsymbol{\tau}_r^{t-1} = [(\mathbf{A}_{n_c} \odot \mathbf{A}_{n_c}^*)^{\mathrm{T}}(\sigma_z^2\mathbf{1}_{M_{\mathrm{all}}^{\mathrm{p}}} + \boldsymbol{\tau}_p^{t-1})^{\circ -1}]^{\circ -1}, \quad (45)$$

with $M_{\mathrm{all}}^{\mathrm{p}} = MN_{\mathrm{v}}^{\mathrm{p}}$, and $\boldsymbol{\mu}_p^{t-1}$ and $\boldsymbol{\tau}_p^{t-1}$ are computed as

$$\boldsymbol{\mu}_p^{t-1} = \mathbf{A}_{n_c}\hat{\mathbf{h}}_{n_c}^{t-1} - \boldsymbol{\tau}_p^{t-1} \odot \mathbf{y}_{n_c}^{t-2}, \quad (46)$$

$$\boldsymbol{\tau}_p^{t-1} = (\mathbf{A}_{n_c} \odot \mathbf{A}_{n_c}^*)\boldsymbol{\tau}_x^{t-1}. \quad (47)$$

The equations above complete the iteration of GAMP algorithm and damping techniques can be applied to ensure the convergence. Specially, for the case with only spatial statistical CSI or without statistical CSI, accurate $\mathbf{r}_{n_c}$ is not feasible as a priori information. In this case, GAMP algorithm can be combined with the expectation-maximization (EM) algorithm. Namely, an estimated $\mathbf{r}_{n_c}^i$ initialized as a vector with the same elements is used for GAMP, and after performing GAMP with $T$ iterations, $\mathbf{r}_{n_c}^i$ can be updated as

$$\mathbf{r}_{n_c}^i = \hat{\mathbf{h}}_{n_c}^{iT} \odot (\hat{\mathbf{h}}_{n_c}^{iT})^* + \boldsymbol{\tau}_x^{iT}, \quad (48)$$

where the superscript $i$ denotes the iteration index of EM. In this case, the algorithm is equivalent to the damped GGAMP-SBL [38]. Then when GAMP converges or the number of iterations exceeds the preset maximum number of iterations, the final estimate of $\mathbf{h}_{n_c}$ is $\hat{\mathbf{h}}_{n_c}$. Denote $\hat{\mathbf{H}}_u^{n_c}$ is the estimate of $\mathbf{H}_u^{n_c}$, which satisfies $\mathrm{vec}([\hat{\mathbf{H}}_0^{n_c}, \hat{\mathbf{H}}_1^{n_c}, \ldots, \hat{\mathbf{H}}_{U_{\mathrm{c}}-1}^{n_c}]) = \hat{\mathbf{h}}_{n_c}$. Then with (38) and (39), the SF domain channel can be estimated.

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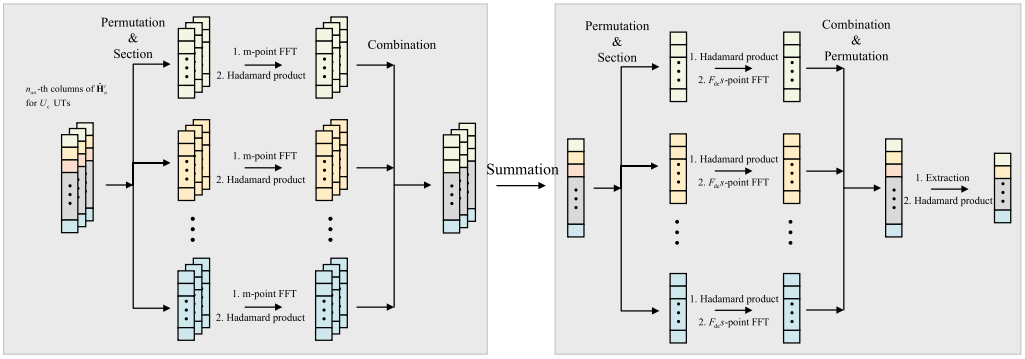



Fig. 2. Illustration of efficient implementation of computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$.

Note that computations involved with $\odot$ and $(\cdot)^{\circ -1}$ are all element-wise. Besides, since the modulus of elements in $\mathbf{A}_{n_c}$ are all 1, $\mathbf{A}_{n_c} \odot \mathbf{A}^*_{n_c}$ is the matrix with all elements being 1. Therefore, matrix products in (45) and (47) are equivalent to summations. Therefore, the computational complexity of GAMP is mainly dominated by computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ in (46) and $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ in (42). From the equivalence between (16) and (18), $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ and $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ can be computed as

$$\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c} = \sum_{u=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}}\hat{\mathbf{H}}^t_u \mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}}\mathbf{X}_u, \tag{49}$$

$$\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c} = \mathrm{vec}([\bar{\mathbf{H}}^t_0, \bar{\mathbf{H}}^t_1, \ldots, \bar{\mathbf{H}}^t_{U_{\mathrm{c}}-1}]), \tag{50}$$

where $\mathrm{vec}([\hat{\mathbf{H}}^t_0, \hat{\mathbf{H}}^t_1, \ldots, \hat{\mathbf{H}}^t_{U_{\mathrm{c}}-1}]) = \hat{\mathbf{h}}^t_{n_c}$, and

$$\bar{\mathbf{H}}^t_u = \mathbf{V}^{\mathrm{H}}_{\mathrm{s}}\mathbf{Y}^t_{n_c}\mathbf{V}^*_{\mathrm{f,p}}\mathbf{X}^*_u,\ u = 0, 1, \ldots, U_{\mathrm{c}}-1, \tag{51}$$

with $\mathrm{vec}(\mathbf{Y}^t_{n_c}) = \mathbf{y}^t_{n_c}$. Considering matrix multiplications with DFT matrices can be implemented by FFT and inverse FFT (IFFT), both the number of complex multiplications of computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ as in (49) and that of computing $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ as in (51) are

$$N_{\mathrm{di}} = M^{\mathrm{p}}_{\mathrm{all}}U_{\mathrm{c}} + \frac{1}{2}N^{\mathrm{p}}_{\mathrm{all}}U_{\mathrm{c}}\log_2\left(\frac{N^{\mathrm{p}}_{\mathrm{all}}}{4}\right), \tag{52}$$

where $N^{\mathrm{p}}_{\mathrm{all}} = N_{\mathrm{an}}N^{\mathrm{p}}_{\mathrm{de}}$, and the number of complex multiplications in FFT is measured as $N/2\log(N/4)$ for $N$-point FFT since no multiplication is needed when FFT size is not greater than 4. Although the matrix multiplications can be implemented by FFT and IFFT, this operation has to be repeated for each UT, which makes the computational complexity increase $U_{\mathrm{c}}$ times.

### B. Efficient Implementation

In this subsection, we propose an efficient implementation for the GAMP algorithm by decomposing the original large-size FFT in $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ and $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ into several small-size FFTs, and then the number of complex multiplications can be significantly decreased.

The efficient implementation of computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ can be divided into two stages, which is derived in Appendix F. The first stage is to compute

$$\hat{\mathbf{H}}^{\mathrm{f}}_u = \hat{\mathbf{H}}^t_u(\mathbf{X}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{D}_{\mathrm{F}}\mathbf{P}_{m,F_{\mathrm{de}}s})^{\mathrm{T}}, \tag{53}$$

where $\mathbf{D}_{\mathrm{F}} = \mathrm{Bdiag}(\mathbf{F}_m, \mathbf{F}_m, \ldots, \mathbf{F}_m) \in \mathbb{C}^{N^{\mathrm{p}}_{\mathrm{de}}\times N^{\mathrm{p}}_{\mathrm{de}}}$, $\mathbf{X}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}} = \mathrm{diag}(\mathbf{x}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}})$, and $\mathbf{P}_{m,F_{\mathrm{de}}s}$ is the permutation matrix constructed by letting $[\mathbf{P}_{m,F_{\mathrm{de}}s}]_{k,F_{\mathrm{de}}sl_1+l_2} = 1$, $k = 0, 1, \ldots, N^{\mathrm{p}}_{\mathrm{de}}-1$, $l_1 = \langle k\rangle_m$, $l_2 = \lfloor k/m\rfloor$, and other elements be 0. In the second stage, $\hat{\mathbf{H}}^{\mathrm{f}}_u$ for different users are summed and $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ are calculated as

$$\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c} = \mathbf{V}_{\mathrm{s}}\left(\sum_{u=0}^{U_{\mathrm{c}}-1}\hat{\mathbf{H}}^{\mathrm{f}}_u\right)(\mathbf{V}^{\mathrm{ef}}_{\mathrm{f}})^{\mathrm{T}}, \tag{54}$$

where $\mathbf{V}^{\mathrm{ef}}_{\mathrm{f}} \triangleq \mathbf{X}^r_{N^{\mathrm{p}}_{\mathrm{v}},N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{I}_{N^{\mathrm{p}}_{\mathrm{v}},N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{P}^{\mathrm{T}}_{F_{\mathrm{de}}s,m}\mathbf{D}_{s,m}\mathbf{P}_{F_{\mathrm{de}}s,m}$. $\mathbf{P}_{F_{\mathrm{de}}s,m}$ is a permutation matrix, and $\mathbf{D}_{s,m}$ can be defined as

$$\begin{aligned}\mathbf{D}_{s,m} &\triangleq \mathbf{P}_{F_{\mathrm{de}}s,m}\mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s}\mathbf{D}^{\mathrm{H}}_{\mathrm{F}}\mathbf{P}^{\mathrm{T}}_{F_{\mathrm{de}}s,m}\\ &= \mathrm{Bdiag}\left(\mathbf{F}_{F_{\mathrm{de}}s}, \mathbf{F}_{F_{\mathrm{de}}s}\widetilde{\mathbf{\Phi}}^1_{N^{\mathrm{p}}_{\mathrm{de}}}, \ldots, \mathbf{F}_{F_{\mathrm{de}}s}\widetilde{\mathbf{\Phi}}^{m-1}_{N^{\mathrm{p}}_{\mathrm{de}}}\right),\end{aligned} \tag{55}$$

where $\widetilde{\mathbf{\Phi}}^k_{N^{\mathrm{p}}_{\mathrm{de}}} \triangleq \mathbf{I}_{F_{\mathrm{de}}s,N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{\Phi}^k_{N^{\mathrm{p}}_{\mathrm{de}}}$. Similarly, $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ can be calculated with the following two stages

$$\mathbf{Y}^{\mathrm{f}} = \mathbf{V}^{\mathrm{H}}_{\mathrm{s}}\mathbf{Y}^t_{n_c}(\mathbf{V}^{\mathrm{ef}}_{\mathrm{f}})^*, \tag{56}$$

$$\bar{\mathbf{H}}^t_u = \mathbf{Y}^{\mathrm{f}}(\mathbf{X}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}}\mathbf{D}_{\mathrm{F}}\mathbf{P}_{m,F_{\mathrm{de}}s})^*. \tag{57}$$

The whole implementation of computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ is displayed in Fig. 2, with the $n_{\mathrm{an}}$-th column of $\hat{\mathbf{H}}^t_u$ for $U_{\mathrm{c}}$ UTs as an example. In this way, the original $N^{\mathrm{p}}_{\mathrm{de}}$-dimensional DFT performed $U_{\mathrm{c}}$ times in the direct implementation is replaced by $m$-dimensional DFT performed $U_{\mathrm{c}}F_{\mathrm{de}}s$ times and $F_{\mathrm{de}}s$-dimensional DFT performed $m$ times in the efficient implementation, by which the computational complexity is reduced. Both the numbers of complex multiplications for calculating $\hat{\mathbf{h}}^t_{n_c}$ with (54) and $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$ with (57) can be measured as

$$\begin{aligned}N_{\mathrm{ef}} = {} & M^{\mathrm{p}}_{\mathrm{all}} + \frac{1}{2}N^{\mathrm{p}}_{\mathrm{all}}\log_2\left(\frac{N_{\mathrm{an}}F_{\mathrm{de}}s}{16}\right)\\ & + \frac{1}{2}N^{\mathrm{p}}_{\mathrm{all}}U_{\mathrm{c}}\log_2\left(\frac{m}{4}\right) + N^{\mathrm{p}}_{\mathrm{all}}(U_{\mathrm{c}}+1).\end{aligned} \tag{58}$$

Then $2N_{\mathrm{di}}$ and $2N_{\mathrm{ef}}$ can be regarded as an approximation of the complex multiplication number in one iteration of the GAMP, since the computational complexity of the GAMP is dominated by computing $\mathbf{A}_{n_c}\hat{\mathbf{h}}^t_{n_c}$ and $\mathbf{A}^{\mathrm{H}}_{n_c}\mathbf{y}^t_{n_c}$. With (52) and (58), the ratio between the number of complex multiplications

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TABLE I
PARAMETER SETTINGS

| Parameter | Value |
|---|---|
| Number of BS antennas $M$ | 128 |
| UT number $U$ | 12, 24, 36, 48 |
| UT number per comb $U_{\mathrm{c}}$ | 1, 2, 3, 4 |
| Center frequency $f_{\mathrm{c}}$ | 5.8 GHz |
| Number of valid subcarriers $N_{\mathrm{v}}$ | 1152 |
| Number of subcarriers per comb $N_{\mathrm{v}}^{\mathrm{P}}$ | 96 |
| Number of combs $N_{\mathrm{c}}$ | 12 |
| Subcarrier spacing $\Delta f$ | 15 kHz |
| Number of subcarriers $N_{\mathrm{sc}}$ | 2048 |
| CP length $N_{\mathrm{cp}}$ | 144 |
| Fine factor $F_{\mathrm{an}}, F_{\mathrm{de}}$ | 2 |

of the efficient implementation and that of the direct implementation is

$$\frac{N_{\mathrm{ef}}}{N_{\mathrm{di}}} = \frac{1}{U_{\mathrm{c}}} + \frac{U_{\mathrm{c}}-1}{U_{\mathrm{c}}} \frac{2F_{\mathrm{an}}F_{\mathrm{de}} + F_{\mathrm{an}}F_{\mathrm{de}}\log_2\left(\frac{m}{4}\right)}{2 + F_{\mathrm{an}}F_{\mathrm{de}}\log_2\left(\frac{N_{\mathrm{all}}^{\mathrm{P}}}{4}\right)}$$
$$\approx \frac{1}{U_{\mathrm{c}}} + \frac{U_{\mathrm{c}}-1}{U_{\mathrm{c}}} \frac{2+\log_2\left(\frac{m}{4}\right)}{\log_2\left(\frac{N_{\mathrm{all}}^{\mathrm{P}}}{4}\right)}. \tag{59}$$

Since $m$ is much smaller than $N_{\mathrm{all}}^{\mathrm{P}}$, the number of complex multiplications of the proposed efficient implementation is significantly decreased, and when $N_{\mathrm{de}}^{\mathrm{P}}$ is fixed, the smaller $m$ leads to fewer complex multiplications needed. Specially, when $M \to \infty$, $N_{\mathrm{all}}^{\mathrm{P}} \to \infty$, we have that $N_{\mathrm{ef}}/N_{\mathrm{di}} \to 1/U_{\mathrm{c}}$, which implies that the efficient implementation can reduce the number of complex multiplications to $1/U_{\mathrm{c}}$ at most. Although we use the GAMP algorithm here, other channel estimation algorithms, such as simplified IGA [39], can also be implemented with this technique.

Moreover, combined with Theorem 3, the GPSPs provide a trade-off between performance and computational complexity. For a fixed $N_{\mathrm{v}}^{\mathrm{P}}$, the performance and computational complexity can be flexibly adjusted by $m$ or $s$.

## V. SIMULATION RESULTS

In this section, simulation results are provided to illustrate the performance of the proposed GPSPs for massive MIMO-OFDM channel acquisition. QuaDRiGa channel model [40] is adopted to generate channels with main parameters summarized in Table I. The channel is normalized as $\|\mathbf{H}_u^{\mathrm{SF}}\|_F^2 = MN_{\mathrm{v}}^{\mathrm{P}}$. The channel is estimated by the GAMP based method in Section IV. Meanwhile, 2D beam domain statistical CSI is obtained from the SF domain CSI by the damped GGAMP-SBL. The normalized MSE (NMSE) defined as $\epsilon_{\mathrm{NMSE}} \triangleq \frac{1}{U}\sum_{u=0}^{U-1} \frac{\|\widehat{\mathbf{H}}_u^{\mathrm{SF}} - \mathbf{H}_u^{\mathrm{SF}}\|_F^2}{\|\mathbf{H}_u^{\mathrm{SF}}\|_F^2}$ is used as the performance metric for channel estimation. We compare the proposed GPSPs with the following pilots.

- ZC: ZC sequences defined as (22), with length $N_{\mathrm{v}}^{\mathrm{P}}$ and $r \in \{0, 1, \ldots, N_{\mathrm{v}}^{\mathrm{P}} - 1\}$.
- APSP: Adjustable phase shift pilots without comb-type pilot structure [14].

Specifically, the transmit power of the pilots for each user is set to 1, the length of both GPSPs and ZC sequences is $N_{\mathrm{v}}^{\mathrm{P}}$ while the length of APSPs is $N_{\mathrm{v}}$ as no comb-type pilot structure is applied. Besides, for GPSPs and ZC sequences, the number of UTs per comb is $U_{\mathrm{c}}$, and the total number for these pilots are identical.

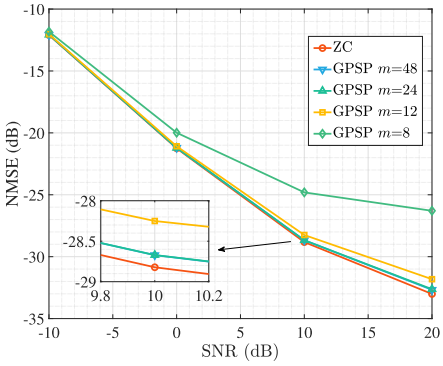



Fig. 3. Channel estimation performance of different pilots without the statistical CSI, $U = 48$.

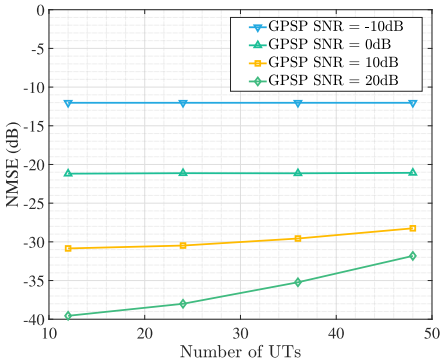



Fig. 4. Channel estimation performance of GPSPs without statistical CSI across different UT numbers, $m = 12$.

Fig. 3 shows the NMSE performance of the channel estimation without the statistical CSI under different pilot configurations.The number of subcarriers per comb is set to 96, which is a typical case in 5G systems [41]. As shown in Fig. 3, when the number of UTs per comb is 4, for low SNR, 8 is enough to maintain the channel estimation performance and the computational complexity is the lowest. For a higher SNR, a larger $m$ such as 12 or 24 is preferred to ensure the channel estimation performance. Meanwhile, as illustrated in Fig. 3, the performance degrades slowly as m decreases when $m > 12$.

Fig. 4 shows the channel estimation performance of GPSPs without statistical CSI across different UT numbers. When the number of UTs is 12, each comb corresponds to a single UT and no inter-user interference exists. This serves as an orthogonal pilot baseline. At low SNR values, the system performance remains stable as the number of UTs increases. In contrast, at high SNR values, performance degrades with a growing number of UTs. However, the resulting estimate can be accurate enough in practical systems.

Fig. 5 shows the NMSE performance of channel estimation with the spatial statistical CSI under different pilot configurations. The pilots with scheduling are scheduled according to

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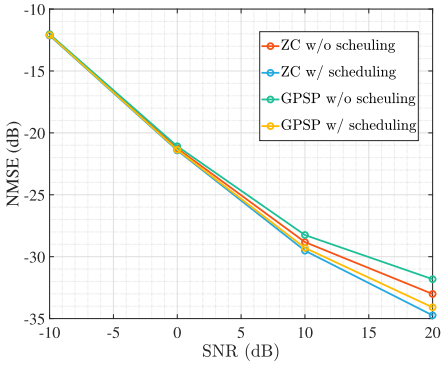



Fig. 5. Channel estimation performance of different pilots with the spatial statistical CSI, $U = 48$, $m = 12$.

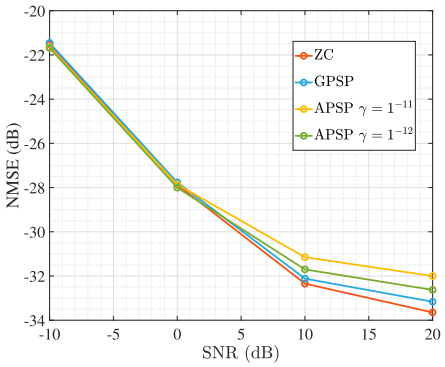



Fig. 6. Channel estimation performance of different pilots with the SF statistical CSI, $U = 96$, $m = 12$.

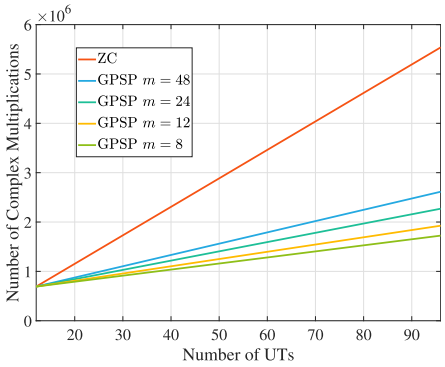



Fig. 7. The number of complex multiplications in one iteration under different pilots, $U = 48$.

Algorithm 1. The results demonstrate when the SNR is low and the channel estimation performance is noise-limited, the NMSE performance with and without scheduling is almost identical. In contrast, when the SNR is high, the inter-user interference becomes dominant, and the proposed scheduling strategy provides a noticeable performance gain.

Fig. 6 shows the NMSE performance of channel estimation with SF statistical CSI under different pilot configurations, where GPSPs and ZC sequences are scheduled by Algorithm 1 while APSPs are scheduled by the pilot phase shift scheduling algorithm in [14]. In this simulation, $m$ is set to 12. Let $U = 96$ and $U_c = 8$. In this case, the performance of GPSPs and ZC sequences are fairly close. Since the pilot schedule algorithm for APSPs just makes the channel overlap function smaller than the preset threshold $\gamma$, which is intended to accelerate the scheduling, the performance of APSPs varies according to the value of $\gamma$. When $\gamma$ is set to $1^{-11}$ and $1^{-12}$, GPSPs and ZC sequences still demonstrate a similar performance, which is better than that of APSPs.

The comparison of the number of complex multiplications required in one iteration under various pilot configurations is illustrated in Fig. 7. The GAMP for ZC is implemented by (49) and (51), while it is implemented by (54) and (57) for GPSPs. Therefore, the numbers of complex multiplications of ZC sequence and GPSPs are in accordance with (52) and (58), respectively. For a fixed number of subcarriers, $N_{\mathrm{v}}^{\mathrm{p}} = sm$, it is observed that a decrease in the value of $m$ results in a reduction in the number of complex multiplications.

## VI. Conclusion

In this paper, we have investigated the non-orthogonal pilot design for massive MIMO-OFDM channel acquisition. With the beam based channel model, we first formulate a signal model with the comb-type pilot structure and derive its simplified equivalent form. Then, we propose the GPSPs and reveal their relevant properties. We analyze the inter-user interference caused by GPSPs, which is negligible when the length of the weight sequence is large enough. Therefore, the feasibility of the proposed GPSPs is proved in the case without the statistical CSI. When the spatial statistical CSI is available, we also propose a pilot scheduling algorithm to mitigate the possible interference. When the SF statistical CSI is available, we derive the conditions to minimize the MSE of channel estimation with GPSPs. Then, utilizing the property of GPSPs, we propose an efficient implementation for channel estimation with much fewer complex multiplications for channel estimation. Simulation results verify the validity of our proposed channel estimation with GPSPs.

## Appendix A
## Proof of Theorem 1

Since the phase shifts are chosen uniformly and $\phi_0$ is set to 0, $\mathbf{\Phi}_{\mathrm{ps}}$ can be expressed as $[\mathbf{f}_0, \mathbf{f}_s, \mathbf{f}_{2s}, \ldots, \mathbf{f}_{N-s}]$, where $s = N/m$ is the spacing length of two adjacent phase shifts. Then we have

$$\mathbf{x}_u = \mathbf{X}_{\mathrm{r}} \begin{bmatrix} \mathbf{F}_m \mathbf{w}_u \\ \mathbf{F}_m \mathbf{w}_u \\ \vdots \\ \mathbf{F}_m \mathbf{w}_u \end{bmatrix} = \sqrt{m} \mathbf{X}_{\mathrm{r}} \begin{bmatrix} \tilde{\mathbf{w}}_u \\ \tilde{\mathbf{w}}_u \\ \vdots \\ \tilde{\mathbf{w}}_u \end{bmatrix}, \tag{60}$$

where $\tilde{\mathbf{w}}_u = \frac{1}{\sqrt{m}} \mathbf{F}_m \mathbf{w}_u$. When $\mathbf{w}_u$ is a ZC sequence of length $m$, $\tilde{\mathbf{w}}_u$ is still a ZC sequence of length $m$ [34], [35]. Since $\mathbf{X}_{\mathrm{r}}$ is with unit modulus, the modulus of $\mathbf{x}_u$ is $\sqrt{m}$.

Then for the DFT of $\mathbf{x}_u$, note that the GPSP $\mathbf{x}_u$ can be rewritten as $\sum_{k=0}^{m-1} [\mathbf{w}_u]_k \mathbf{D}_{\phi_k} \mathbf{x}_{\mathrm{r}}$. Then we have

$$\tilde{\mathbf{x}}_u = \sum_{k=0}^{m-1} [\mathbf{w}_u]_k \mathbf{\Pi}_N^{-\phi_k} \tilde{\mathbf{x}}_{\mathrm{r}}. \tag{61}$$

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Therefore, we have

$$[\tilde{\mathbf{x}}_u]_n^{\mathrm{H}}[\tilde{\mathbf{x}}_u]_n = \sum_{k=0}^{m-1}\sum_{l=0}^{m-1}[\mathbf{w}_u]_k^{\mathrm{H}}[\tilde{\mathbf{x}}_{\mathrm{r}}]_{n-\phi_k}^{\mathrm{H}}[\mathbf{w}_u]_l[\tilde{\mathbf{x}}_{\mathrm{r}}]_{n-\phi_l}. \quad (62)$$

The periodicity of ZC sequences allows the change of variable $l \to l+k$, and with $\phi_0 = 0$, we have that $\phi_k = ks$. Thus, for the case that $N$ and $m$ are both even, we have

$$[\tilde{\mathbf{x}}_u]_n^{\mathrm{H}}[\tilde{\mathbf{x}}_u]_n = \sum_{k=0}^{m-1} e^{-\bar{\jmath}\pi q\frac{k^2}{m}} e^{-\bar{\jmath}\pi\tilde{r}\frac{sk^2-2nk}{m}} \sum_{l=0}^{m-1} e^{-\bar{\jmath}2\pi(q+s\tilde{r})\frac{kl}{m}}. \quad (63)$$

Since $q+s\tilde{r}$ is relatively prime to $m$, $\langle(q+s\tilde{r})k\rangle_m = 0$ if and only if $k$ is 0 for $k \in [0,1,\ldots,m-1]$. Thus, the second summation in (63) is 0 unless $k=0$. Then it can be easily shown that $|[\tilde{\mathbf{x}}_u]_n| = \sqrt{m}$ for all $n$. In this way, the proof can be completed for the case with $N$ and $m$ both even. For odd $N$ or $m$, the expression of ZC sequence $\mathbf{w}_u$ or $\tilde{\mathbf{x}}_{\mathrm{r}}$ is different, and therefore $[\tilde{\mathbf{x}}_u]_n^{\mathrm{H}}[\tilde{\mathbf{x}}_u]_n$ changes from (63), but the proof is similar. This completes the proof.

## Appendix B
## Proof of Theorem 2

For the autocorrelation property of $\tilde{\mathbf{x}}_u$, since $|\frac{1}{\sqrt{N}}\mathbf{F}_N\tilde{\mathbf{x}}_u|^2 = |\mathbf{x}_u|^2 = m$ as proved in Theorem 1, where the absolute value sign $|\cdot|$ means the absolute value for every element of a vector, we have $R_a[p] = mN\delta[p]$ due to the circular correlation property of DFT. Then for the crosscorrelation property of $\tilde{\mathbf{x}}_u$, substituting (61) into (26), we have

$$R_c^{u,u'}[p] = \sum_{k=0}^{m-1}\sum_{l=0}^{m-1}[\mathbf{w}_{u'}]_k^{\mathrm{H}}[\mathbf{w}_u]_l\tilde{\mathbf{x}}_{\mathrm{r}}^{\mathrm{H}}\mathbf{\Pi}_N^{\phi_k-\phi_l+p}\tilde{\mathbf{x}}_{\mathrm{r}}. \quad (64)$$

When $\langle p\rangle_s \neq 0$, $\langle\phi_k-\phi_l+p\rangle_N \neq 0$. Then, utilizing the autocorrelation property of a ZC sequence, $\tilde{\mathbf{x}}_{\mathrm{r}}^{\mathrm{H}}\mathbf{\Pi}_N^{\phi_k-\phi_l+p}\tilde{\mathbf{x}}_{\mathrm{r}} = 0$ and thus $R_c[p]=0$. When $\langle p\rangle_s = 0$, there exist $k$, $l$ such that $\langle\phi_k-\phi_l+p\rangle_N = 0$. In this case, we have $R_c^{u,u'}[p] = N\mathbf{w}_{u'}^{\mathrm{H}}\mathbf{\Pi}_N^{\frac{p}{s}}\mathbf{w}_u$. Then utilizing the crosscorrelation property of ZC sequences, we have $|R_c^{u,u'}[p]| = \sqrt{m}N$ when $\langle p\rangle_s = 0$. This completes the proof.

## Appendix C
## Derivation of (30)

The derivation of (30) is as below

$$\begin{aligned}\mathbf{X}_u\mathbf{V}_{\mathrm{f,p}} &\overset{(a)}{=} \mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{X}^{q_u}_{m,N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}\\ &\overset{(b)}{=} \mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{X}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}\\ &\overset{(c)}{=} \mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}})\\ &= \mathbf{X}^r_{N_{\mathrm{v}}^{\mathrm{p}}}\widetilde{\mathbf{V}}_{\mathrm{f,p}}\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}),\end{aligned} \quad (65)$$

where (a) is from (15), (b) is from $\mathbf{X}^{q_u}_{m,N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}} = \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{X}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}$, (c) is from that circulant matrices can be diagonalized by discrete Fourier matrices [42]. Since $\mathbf{X}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}} = \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}})$, we have $\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}) = \frac{1}{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}^{\mathrm{H}}_{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{X}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}$ and thus $\widetilde{\mathbf{x}}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}} = \frac{1}{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}^{\mathrm{H}}_{N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{x}^{q_u}_{m,N_{\mathrm{de}}^{\mathrm{p}}}$.

## Appendix D
## Proof of Theorem 3

Since the number of propagation paths is limited, the number of non-zero elements in $[\mathbf{H}^{n_{\mathrm{c}}}_{u'}]_{n_{\mathrm{an}},:}$ is also limited. Denote the indexes of non-zero elements in $[\mathbf{H}^{n_{\mathrm{c}}}_{u'}]_{n_{\mathrm{an}},:}$ as the set $\mathbb{P}_{u,n_{\mathrm{an}}}$, then the interference can be rewritten as

$$\begin{aligned}&[\mathbf{I}^{\mathrm{B}}_{n_{\mathrm{c}},u}]_{n_{\mathrm{an}},n_{\mathrm{de}}}\\ &= \sum_{u'\neq u}\sum_{p\in\mathbb{P}_{u',n_{\mathrm{an}}}}[\mathbf{H}^{n_{\mathrm{c}}}_{u'}]_{n_{\mathrm{an}},p}[(\circ(\widetilde{\mathbf{x}}^{q_u,q_{u'}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}))^{\mathrm{T}}]_{p,n_{\mathrm{de}}},\end{aligned} \quad (66)$$

Since the magnitudes of elements of $\mathbf{H}^{n_{\mathrm{c}}}_u$ are upper bounded, denote the maximum of them as $H_{max}$. Then utilizing Theorem 2, an upper bound of the magnitude of $[\mathbf{I}^{\mathrm{B}}_{n_{\mathrm{c}},u}]_{n_{\mathrm{an}},n_{\mathrm{de}}}$ can be rewritten as

$$|[\mathbf{I}^{\mathrm{B}}_{n_{\mathrm{c}},u}]_{n_{\mathrm{an}},n_{\mathrm{de}}}| \leq \frac{1}{\sqrt{m}}H_{max}P_{n_{\mathrm{an}}}, \quad (67)$$

where $P_{n_{\mathrm{an}}} = \sum_{u'\neq u}P_{u',n_{\mathrm{an}}}$ and $P_{u,n_{\mathrm{an}}} = |\mathbb{P}_{u,n_{\mathrm{an}}}|$ represents the number of elements in set $\mathbb{P}_{u,n_{\mathrm{an}}}$. When $m$ goes to infinity, we can derive the limit as follows,

$$\lim_{m\to\infty}|[\mathbf{I}^{\mathrm{B}}_{n_{\mathrm{c}},u}]_{n_{\mathrm{an}},n_{\mathrm{de}}}| \leq \lim_{m\to\infty}\frac{1}{\sqrt{m}}H_{max}P_{n_{\mathrm{an}}} = 0. \quad (68)$$

Therefore, we have

$$\lim_{m\to\infty}[\mathbf{I}^{\mathrm{B}}_{n_{\mathrm{c}},u}]_{n_{\mathrm{an}},n_{\mathrm{de}}} = 0, \quad (69)$$

for all $n_{\mathrm{an}}$ and $n_{\mathrm{de}}$. This completes the proof.

## Appendix E
## Proof of Theorem 5

Before proving Theorem 5, we present the following Lemma 1 and Lemma 2 for two conditions, respectively.

*Lemma 1:* When $M\to\infty$ and for $\forall u\neq u'$ in the $n_{\mathrm{c}}$-th comb $\eta^{\mathrm{an}}_{n_{\mathrm{c}},u,u'} = 0$, we have

$$\lim_{M\to\infty}\frac{\|\bar{\mathbf{X}}_u\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}\bar{\mathbf{X}}^{\mathrm{H}}_u\bar{\mathbf{X}}_{u'}\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u'}\|_F}{\|\bar{\mathbf{X}}_u\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}\bar{\mathbf{X}}^{\mathrm{H}}_u\bar{\mathbf{X}}_u\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}\|_F} = 0. \quad (70)$$

*Proof:* Since $\mathbf{h}^{\mathrm{SF}}_{n_{\mathrm{c}},u} = \mathrm{vec}(\mathbf{V}_{\mathrm{s}}\mathbf{H}^{n_{\mathrm{c}}}_u\mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}})$, we have $\mathbf{h}^{\mathrm{SF}}_{n_{\mathrm{c}},u} = \mathbf{V}\mathrm{vec}(\mathbf{H}^{n_{\mathrm{c}}}_u)$ with $\mathbf{V} = \mathbf{V}_{\mathrm{f,p}}\otimes\mathbf{V}_{\mathrm{s}}$, and then $\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} = \mathbf{V}\mathbf{R}^{n_{\mathrm{c}}}_u\mathbf{V}^{\mathrm{H}}$ where $\mathbf{R}^{n_{\mathrm{c}}}_u = \mathbb{E}\{\mathrm{vec}(\mathbf{H}^{n_{\mathrm{c}}}_u)\mathrm{vec}^{\mathrm{H}}(\mathbf{H}^{n_{\mathrm{c}}}_u)\}$ that is diagonal as mentioned before. Thus,

$$\bar{\mathbf{X}}_u\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}\bar{\mathbf{X}}^{\mathrm{H}}_u\bar{\mathbf{X}}_{u'}\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u'} = \bar{\mathbf{X}}_u\mathbf{V}\mathbf{\Phi}_{u,u'}\mathbf{V}^{\mathrm{H}}, \quad (71)$$

where $\mathbf{\Phi}_{u,u'}$ is defined as $\mathbf{\Phi}_{u,u'} = \mathbf{R}^{n_{\mathrm{c}}}_u\mathbf{V}^{\mathrm{H}}\bar{\mathbf{X}}^{\mathrm{H}}_u\bar{\mathbf{X}}_{u'}\mathbf{V}\mathbf{R}^{n_{\mathrm{c}}}_{u'}$. Then $[\mathbf{\Phi}_{u,u'}]_{a,b}$ can be expressed as

$$\begin{aligned}[\mathbf{\Phi}_{u,u'}]_{a,b} = [\mathbf{R}^{n_{\mathrm{c}}}_u]_{a,a}[\mathbf{R}^{n_{\mathrm{c}}}_{u'}]_{b,b}\times\alpha_M\left(\frac{n'_{\mathrm{an}}-n_{\mathrm{an}}}{F_{\mathrm{an}}}\right)\\ \times\underbrace{\sum_{k=0}^{N_{\mathrm{v}}^{\mathrm{p}}-1}[\mathbf{x}^{q_u}_{m,N_{\mathrm{v}}^{\mathrm{p}}}]^*_k[\mathbf{x}^{q_{u'}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}]_k e^{-\bar{\jmath}2\pi k\frac{n'_{\mathrm{de}}-n_{\mathrm{de}}}{N_{\mathrm{de}}^{\mathrm{p}}}}}_{\phi_{\mathrm{de}}},\end{aligned} \quad (72)$$

where $[\mathbf{x}^{q_u}_{m,N_{\mathrm{v}}^{\mathrm{p}}}]_n = [\mathbf{x}^{q_u}_m]_{\langle n\rangle_m}$ with $n = 0,1,2,\ldots,N_{\mathrm{v}}^{\mathrm{p}}-1$, $n_{\mathrm{an}} = \langle a\rangle_{N_{\mathrm{an}}}$, $n_{\mathrm{de}} = \lfloor a/N_{\mathrm{an}}\rfloor$, $n'_{\mathrm{an}} = \langle b\rangle_{N_{\mathrm{an}}}$, $n'_{\mathrm{de}} = \lfloor b/N_{\mathrm{an}}\rfloor$, and $\alpha_M(x)\triangleq\sum_{a=0}^{M-1}e^{-\bar{\jmath}2\pi\frac{ax}{M}}$.

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When $[\mathbf{R}_u^{n_c}]_{a,a}[\mathbf{R}_{u'}^{n_c}]_{b,b} = 0$, we have $[\mathbf{\Phi}_{u,u'}]_{a,b} = 0$. When $[\mathbf{R}_u^{n_c}]_{a,a}[\mathbf{R}_{u'}^{n_c}]_{b,b} \neq 0$, due to $\eta_{n_c,u,u'}^{\mathrm{an}} = 0$, we have $n'_{\mathrm{an}} \neq n_{\mathrm{an}}$. Since $|\alpha_M(x)| = |\frac{\sin(\pi x)}{\sin(\pi x/M)}|$, as $M \to \infty$, $\alpha_M(x)$ tends to $\delta(x)$ [33]. As $n'_{\mathrm{an}} \neq n_{\mathrm{an}}$, we have $\lim_{M\to\infty} \alpha_M\left(\frac{n'_{\mathrm{an}}-n_{\mathrm{an}}}{F_{\mathrm{an}}}\right) = \delta\left(\frac{n'_{\mathrm{an}}-n_{\mathrm{an}}}{F_{\mathrm{an}}}\right) = 0$. Then since the gain of each propagation path is fixed, the maximum of $[\mathbf{R}_u^{n_c}]_{a,a}[\mathbf{R}_{u'}^{n_c}]_{b,b}$ is independent with $M$ and $|\phi_{\mathrm{de}}|$ is no greater than $N_{\mathrm{v}}^{\mathrm{p}}$, thus we have $\lim_{M\to\infty}[\mathbf{\Phi}_{u,u'}]_{a,b} = 0$.

Denote $\mathcal{P}_u = \{n|[\mathbf{R}_u^{n_c}]_{n,n} \neq 0\}$. As the number of propagation paths is fixed, for any number $M$ of antennas, $\mathcal{P}_u$ only has fixed number of elements while the elements of $\mathcal{P}_u$ may change with $M$. Then for $a \notin \mathcal{P}_u$ or $b \notin \mathcal{P}_{u'}$, we have $[\mathbf{\Phi}_{u,u'}]_{a,b} = 0$. Thus, we have

$$|[\mathbf{V}\mathbf{\Phi}_{u,u'}\mathbf{V}^{\mathrm{H}}]_{n,k}| = |\sum_{a\in\mathcal{P}_u}\sum_{b\in\mathcal{P}_{u'}}[\mathbf{V}]_{n,a}[\mathbf{\Phi}_{u,u'}]_{a,b}[\mathbf{V}^{\mathrm{H}}]_{b,k}|. \quad (73)$$

Since for all $n$ and $k$, we have $|[\mathbf{V}]_{n,k}| = 1$, and thus we have

$$|[\mathbf{V}\mathbf{\Phi}_{u,u'}\mathbf{V}^{\mathrm{H}}]_{n,k}| \leq \sum_{a\in\mathcal{P}_u}\sum_{b\in\mathcal{P}_{u'}}|[\mathbf{\Phi}_{u,u'}]_{a,b}| \leq P_u P_{u'}\phi_{u,u'}. \quad (74)$$

where $\phi_{u,u'}$ is the maximal absolute value of $[\mathbf{\Phi}_{u,u'}]_{a,b}$ for $a \in \mathcal{P}_u$ and $b \in \mathcal{P}_{u'}$, and $P_u$ is the number of elements in $\mathcal{P}_u$.

From the above argument we have $\phi_{u,u'}$ is independent of $M$ and $\lim_{M\to\infty}\phi_{u,u'} = 0$. Thus, according to (71), with $\|\mathbf{A}\|_F^2 = \sum_{i,j}[\mathbf{A}]_{i,j}^2$, we have

$$\lim_{M\to\infty}\|\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2 \leq \lim_{M\to\infty}(MN_{\mathrm{v}}^{\mathrm{p}}P_uP_{u'}\phi_{u,u'})^2. \quad (75)$$

Then since $\bar{\mathbf{X}}_u$ is diagonal and the modulus of elements in diagonal is 1, we have $\|\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2 = \|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2$. Similarly, as $\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u = \mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}}$, we have $\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F = \|\mathbf{R}_{n_c,u}^{\mathrm{SF}}\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F$. Then, with $\|\mathbf{A}\|_F^2 = \mathrm{tr}(\mathbf{A}^{\mathrm{H}}\mathbf{A})$ and $\mathbf{R}_{n_c,u}^{\mathrm{SF}} = \mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}$ Hermitian, we have

$$\|\mathbf{R}_{n_c,u}^{\mathrm{SF}}\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F^2 = \mathrm{tr}((\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}})^4). \quad (76)$$

It is easy to verify that for all $n$, we have $[\mathbf{V}^{\mathrm{H}}\mathbf{V}]_{n,n} = MN_{\mathrm{v}}^{\mathrm{p}}$. Then since $\mathbf{R}_u^{n_c}$ is diagonal, we have

$$\mathrm{tr}(\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}) = \mathrm{tr}(\mathbf{V}^{\mathrm{H}}\mathbf{V}\mathbf{R}_u^{n_c}) = MN_{\mathrm{v}}^{\mathrm{p}}\mathrm{tr}(\mathbf{R}_u^{n_c}). \quad (77)$$

Since $\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}$ is Hermitian and $\mathbf{R}_u^{n_c}$ is positive semi-definite, $\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}$ is positive semi-definite, and the number of positive eigenvalues of $\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}$ is no more than that of $\mathbf{R}_u^{n_c}$. Due to the fixed number of propagation paths, the number of positive eigenvalues of $\mathbf{R}_u^{n_c}$ is fixed. Therefore, the number of positive eigenvalues of $\mathrm{tr}(\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}})$ is fixed. Denote the number of positive eigenvalues of $\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}$ as $N_{\mathrm{pos}}$, and its positive eigenvalues as $\lambda_n$ with $n = 1, 2, \ldots, N_{\mathrm{pos}}$. Then, $\mathrm{tr}(\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}) = \sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n$ and $\mathrm{tr}((\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}})^4) = \sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n^4$. Thus, with Holder's inequality, we have $\left(\sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n^4\right)\left(\sum_{n=1}^{N_{\mathrm{pos}}}1\right)^3 \geq \left(\sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n\right)^4$. Therefore, $\mathrm{tr}((\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}})^4) \geq \left(\sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n\right)^4/(N_{\mathrm{pos}})^3$. Then combined with $\mathrm{tr}(\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}}) = \sum_{n=1}^{N_{\mathrm{pos}}}\lambda_n$, (76) and (77), we have

$$\|\mathbf{R}_{n_c,u}^{\mathrm{SF}}\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F^2 = \mathrm{tr}((\mathbf{V}\mathbf{R}_u^{n_c}\mathbf{V}^{\mathrm{H}})^4) \geq M^4(N_{\mathrm{v}}^{\mathrm{p}})^4c_u, \quad (78)$$

where $c_u = (\mathrm{tr}(\mathbf{R}_u^{n_c}))^4/(N_{\mathrm{pos}})^3$. Since the number of propagation paths is fixed and the maximal gain of all the paths is fixed, $\mathrm{tr}(\mathbf{R}_u^{n_c})$ is fixed. Thus, $c_u$ is fixed. Then we have

$$\lim_{M\to\infty}\frac{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2}{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F^2} \leq \lim_{M\to\infty}\frac{(MN_{\mathrm{v}}^{\mathrm{p}}P_uP_{u'}\phi_{u,u'})^2}{M^4(N_{\mathrm{v}}^{\mathrm{p}})^4c_u} = 0. \quad (79)$$

Thus, (1) holds. This completes the proof of Lemma 1. ■

*Lemma 2:* When $m \to \infty$, for any $u \neq u'$, we have

$$\lim_{m\to\infty}\frac{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F}{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F} = 0. \quad (80)$$

*Proof:* As $N_{\mathrm{de}}^{\mathrm{p}} = F_{\mathrm{de}}N_{\mathrm{v}}^{\mathrm{p}}$, $\phi_{\mathrm{de}}$ in (72) can be rewritten as

$$\phi_{\mathrm{de}} = \sum_{k=0}^{N_{\mathrm{v}}^{\mathrm{p}}-1}[\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u}]_k e^{-\bar{\jmath}2\pi k\frac{n'_{\mathrm{de}}-n_{\mathrm{de}}}{N_{\mathrm{de}}^{\mathrm{p}}}(1-F_{\mathrm{de}})}e^{-\bar{\jmath}2\pi k\frac{n'_{\mathrm{de}}-n_{\mathrm{de}}}{N_{\mathrm{v}}^{\mathrm{p}}}}, \quad (81)$$

Define vector $\boldsymbol{\alpha}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}$ with its $k$-th element $[\boldsymbol{\alpha}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}]_k = \exp\left(-\bar{\jmath}2\pi k\frac{n'_{\mathrm{de}}-n_{\mathrm{de}}}{N_{\mathrm{de}}^{\mathrm{p}}}(1-F_{\mathrm{de}})\right)$. Then, $\phi_{\mathrm{de}}$ is the $(n'_{\mathrm{de}}-n_{\mathrm{de}})$-th element of the $N_{\mathrm{v}}^{\mathrm{p}}$-dimensional DFT of the multiplication of $\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u}$ and $\boldsymbol{\alpha}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}$, which equals the $(n'_{\mathrm{de}}-n_{\mathrm{de}})$-th element of the circular convolution of their respective DFTs.

Let the DFT of $\boldsymbol{\alpha}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}$ be $\tilde{\boldsymbol{\alpha}}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}$, which satisfies $[\tilde{\boldsymbol{\alpha}}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}]_n = \alpha_{N_{\mathrm{v}}^{\mathrm{p}}}(n + (n'_{\mathrm{de}} - n_{\mathrm{de}})(1/F_{\mathrm{de}} - 1))$ with $\alpha_{N_{\mathrm{v}}^{\mathrm{p}}}(x)$ defined as $\alpha_M(x)$ in the proof of Lemma 1. Then similar to the proof in Lemma 1, as $N_{\mathrm{v}}^{\mathrm{p}} = sm$, we have $\lim_{m\to\infty}[\tilde{\boldsymbol{\alpha}}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}]_n = \lim_{N_{\mathrm{v}}^{\mathrm{p}}\to\infty}\alpha_{N_{\mathrm{v}}^{\mathrm{p}}}(n + (n'_{\mathrm{de}} - n_{\mathrm{de}})(1/F_{\mathrm{de}} - 1)) = \delta(n + (n'_{\mathrm{de}} - n_{\mathrm{de}})(1/F_{\mathrm{de}} - 1))$. Note that when $n + (n'_{\mathrm{de}} - n_{\mathrm{de}})(1/F_{\mathrm{de}} - 1) = 0$, we have $\lim_{N_{\mathrm{v}}^{\mathrm{p}}\to\infty}\frac{1}{N_{\mathrm{v}}^{\mathrm{p}}}\alpha_{N_{\mathrm{v}}^{\mathrm{p}}} = \lim_{N_{\mathrm{v}}^{\mathrm{p}}\to\infty}\frac{1}{N_{\mathrm{v}}^{\mathrm{p}}}\sum_{k=0}^{N_{\mathrm{v}}^{\mathrm{p}}-1}1 = 1$. Therefore, $\lim_{N_{\mathrm{v}}^{\mathrm{p}}\to\infty}\frac{1}{N_{\mathrm{v}}^{\mathrm{p}}}\alpha_{N_{\mathrm{v}}^{\mathrm{p}}}$ is the unit sample sequence $\delta[n]$. Then, we have

$$\lim_{m\to\infty}\phi_{\mathrm{de}} = \lim_{m\to\infty}\left[\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u} \oplus_{N_{\mathrm{v}}^{\mathrm{p}}} \frac{1}{N_{\mathrm{v}}^{\mathrm{p}}}\tilde{\boldsymbol{\alpha}}_{n'_{\mathrm{de}}-n_{\mathrm{de}}}\right]_{n'_{\mathrm{de}}-n_{\mathrm{de}}} = \lim_{m\to\infty}[\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u}]_{\langle\frac{1}{F_{\mathrm{de}}}(n'_{\mathrm{de}}-n_{\mathrm{de}})\rangle_{N_{\mathrm{v}}^{\mathrm{p}}}}, \quad (82)$$

where $\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u} = \mathbf{F}_{N_{\mathrm{v}}^{\mathrm{p}}}\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u}$ and $\oplus_N$ represents circular convolution with $N$ point. According to Theorem 2, we have $|[\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_{u'}-q_u}]_{\langle\frac{1}{F_{\mathrm{de}}}(n'_{\mathrm{de}}-n_{\mathrm{de}})\rangle_{N_{\mathrm{v}}^{\mathrm{p}}}}| \leq \frac{1}{\sqrt{m}}|[\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{0}]_0| = s\sqrt{m}$ as $[\bar{\mathbf{x}}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{0}]_0 = N_{\mathrm{v}}^{\mathrm{p}}$. Different from the proof in Lemma 1, when Condition 1) does not hold, $n'_{\mathrm{an}} \neq n_{\mathrm{an}}$ may not hold. However, for a fixed $M$, we have $|\alpha_M\left(\frac{n'_{\mathrm{an}}-n_{\mathrm{an}}}{F_{\mathrm{an}}}\right)| \leq M$. Thus, we have

$$\lim_{m\to\infty}|[\mathbf{\Phi}_{u,u'}]_{a,b}| \leq \lim_{m\to\infty}\beta_{u,u'}Ms\sqrt{m}, \quad (83)$$

where $[\mathbf{\Phi}_{u,u'}]_{a,b}$ is defined in (72), and $\beta_{u,u'}$ is the maximal $[\mathbf{R}_u^{n_c}]_{a,a}[\mathbf{R}_{u'}^{n_c}]_{b,b}$ for all $a$ and $b$. Similar to the proof of Lemma 1, due to the fixed number and maximal gain of propagation paths, with $N_{\mathrm{v}}^{\mathrm{p}} = sm$, we have

$$\lim_{m\to\infty}\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2 \leq \lim_{m\to\infty}M^4s^4m^3P_u^2P_{u'}^2\beta_{u,u'}^2, \quad (84)$$

and $\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F^2 \geq M^4s^4m^4c_u$. Then we have

$$\lim_{m\to\infty}\frac{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_{u'}\mathbf{R}_{n_c,u'}^{\mathrm{SF}}\|_F^2}{\|\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\bar{\mathbf{X}}_u^{\mathrm{H}}\bar{\mathbf{X}}_u\mathbf{R}_{n_c,u}^{\mathrm{SF}}\|_F^2} \leq 0. \quad (85)$$

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Thus, (2) holds. This completes the proof of Lemma 2. ■

Since the number of UTs is fixed, when $M$ is sufficiently large and for $\forall u \neq u'$ in the $n_{\mathrm{c}}$-th comb $\eta^{\mathrm{an}}_{n_{\mathrm{c}},u,u'} = 0$, with Lemma 1, $\bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \bar{\mathbf{X}}_{u'} \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u'}$ is negligible compared to $\bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}$ in the sense of relative Frobenius norm. Thus, we have

$$\mathbf{C}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \approx \widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}, \tag{86}$$

where $\mathbf{C}_{n_{\mathrm{c}}} = \sum_{u=0}^{U_{\mathrm{c}}-1} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}}$ and $\widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} = \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}}$. Then, we have

$$\widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} = \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} (\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}}). \tag{87}$$

Since $\mathbf{C}_{n_{\mathrm{c}}}$, $\widetilde{\mathbf{C}}_{n_{\mathrm{c}},u}$ and $(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}})$ are positive definite, they are invertible. Thus, with (86) and (87), we have

$$\bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} (\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}})^{-1} \approx \mathbf{C}^{-1}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}. \tag{88}$$

$$\bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} (\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}})^{-1} = \widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}. \tag{89}$$

Then with (88) and (89), we have $\mathbf{C}^{-1}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \approx \widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}$. Thus, we have

$$\begin{aligned} &\mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \mathbf{C}^{-1}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}) \\ &\approx \mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}). \end{aligned} \tag{90}$$

Similarly, when $m$ is sufficiently large, with Lemma 2, the relation above also holds. Then for each term of the summation in (40), with the relation (90), we have

$$\begin{aligned} \epsilon &= \sum_{u=0}^{U_{\mathrm{c}}-1} \mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \mathbf{C}^{-1}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}) \\ &\approx \sum_{u=0}^{U_{\mathrm{c}}-1} \mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}) = \epsilon_{\min}. \end{aligned} \tag{91}$$

Since $\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} = \mathbb{E}\{\mathbf{h}^{\mathrm{SF}}_{n_{\mathrm{c}},u} (\mathbf{h}^{\mathrm{SF}}_{n_{\mathrm{c}},u})^{\mathrm{H}}\}$, $\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}$ is positive semi-definite, which is denoted by $\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \succcurlyeq 0$. Then we have $\widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} = (\bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u + \sigma_z^2 \mathbf{I}_{MN^{\mathrm{p}}_{\mathrm{v}}})$ is positive definite, which is denoted as $\widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} \succ 0$. Similarly, we have $\mathbf{C}_{n_{\mathrm{c}}} \succ 0$ and $\mathbf{C}_{n_{\mathrm{c}}} - \widetilde{\mathbf{C}}_{n_{\mathrm{c}},u} \succcurlyeq 0$. Then according to [43], we have $\widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} - \mathbf{C}^{-1}_{n_{\mathrm{c}}} \succcurlyeq 0$, and thus $\mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \mathbf{C}^{-1}_{n_{\mathrm{c}}} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u}) \geq \mathrm{tr}(\mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} - \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}^{\mathrm{H}}_u \widetilde{\mathbf{C}}^{-1}_{n_{\mathrm{c}},u} \bar{\mathbf{X}}_u \mathbf{R}^{\mathrm{SF}}_{n_{\mathrm{c}},u})$ for any $u$. Therefore, $\epsilon_{\min}$ is a lower bound of $\epsilon$, and it can be approached when Condition 1) or Condition 2) is satisfied as proved in (91). This completes the proof of Theorem 5.

## Appendix F
## Derivation of Efficient Implementation

Substituting (30) into (49), the computation of $\mathbf{A}_{n_{\mathrm{c}}} \hat{\mathbf{h}}^t_{n_{\mathrm{c}}}$ can be rewritten as

$$\mathbf{A}_{n_{\mathrm{c}}} \hat{\mathbf{h}}^t_{n_{\mathrm{c}}} = \mathbf{V}_{\mathrm{s}} \left( \sum_{u=0}^{U_{\mathrm{c}}-1} \hat{\mathbf{H}}^t_u (\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}}))^{\mathrm{T}} \right) \mathbf{V}^{\mathrm{T}}_{\mathrm{f,p}} \mathbf{X}^r_{N^{\mathrm{p}}_{\mathrm{v}}}. \tag{92}$$

The non-zero elements in circulant matrix $\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}})$ can be gathered as a block diagonal matrix by permutation matrix $\mathbf{P}_{m,F_{\mathrm{de}}s}$, which can be constructed by letting $[\mathbf{P}_{m,F_{\mathrm{de}}s}]_{k,F_{\mathrm{de}}sl_1+l_2} = 1$, $k = 0,1,\ldots,N^{\mathrm{p}}_{\mathrm{de}}-1$, $l_1 = \langle k \rangle_m$, $l_2 = \lfloor k/m \rfloor$ and the other elements be 0. Thus, we have

$$(\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}}))^{\mathrm{T}} = \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{T}}_{\mathrm{c}} \mathbf{P}_{m,F_{\mathrm{de}}s}, \tag{93}$$

where the block diagonal matrix can be expressed as

$$\mathbf{D}_{\mathrm{c}} = \mathrm{Bdiag} \underbrace{(\circ(\widetilde{\mathbf{x}}^{q_u}_m), \circ(\widetilde{\mathbf{x}}^{q_u}_m), \ldots, \circ(\widetilde{\mathbf{x}}^{q_u}_m))}_{\text{repeat } F_{\mathrm{de}}s \text{ times}}. \tag{94}$$

Note that all the blocks in $\mathbf{D}_{\mathrm{c}}$ are the same circulant matrix $\circ(\widetilde{\mathbf{x}}^{q_u}_m)$. As circulant matrices can be diagonalized by DFT matrices [42], $\circ(\widetilde{\mathbf{x}}^{q}_m)$ can be decomposed as

$$\circ(\widetilde{\mathbf{x}}^{q_u}_m) = \frac{1}{m} \mathbf{F}^{\mathrm{H}}_m \mathrm{diag}(\mathbf{x}^{q_u}_m) \mathbf{F}_m. \tag{95}$$

Substituting (94) and (95) into (93), we have

$$\circ(\widetilde{\mathbf{x}}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}}) = \frac{1}{m} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} \mathbf{X}^q_{m,N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{D}_{\mathrm{F}} \mathbf{P}_{m,F_{\mathrm{de}}s}, \tag{96}$$

where $\mathbf{D}_{\mathrm{F}} = \mathrm{Bdiag}(\mathbf{F}_m, \mathbf{F}_m, \ldots, \mathbf{F}_m)$ and $\mathbf{F}_m$ repeats $F_{\mathrm{de}}s$ times. Then (92) can be rewritten as

$$\begin{aligned} \mathbf{A}_{n_{\mathrm{c}}} \hat{\mathbf{h}}^t_{n_{\mathrm{c}}} = &\frac{1}{m} \mathbf{V}_{\mathrm{s}} \left( \sum_{u=0}^{U_{\mathrm{c}}-1} \hat{\mathbf{H}}_u \left( \mathbf{X}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{D}_{\mathrm{F}} \mathbf{P}_{m,F_{\mathrm{de}}s} \right)^{\mathrm{T}} \right) \\ &\times \left( \mathbf{X}^r_{N^{\mathrm{p}}_{\mathrm{v}}} \mathbf{V}_{\mathrm{f,p}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} \right)^{\mathrm{T}}. \end{aligned} \tag{97}$$

Thus, utilizing (15), we have

$$\mathbf{V}_{\mathrm{f,p}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} = \mathbf{I}_{N^{\mathrm{p}}_{\mathrm{v}},N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}}, \tag{98}$$

where $\mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s}$ can be rewritten as

$$\mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} = \left[ \mathbf{F}^{F_{\mathrm{de}}s}_m, \mathbf{\Phi}^1_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{F}^{F_{\mathrm{de}}s}_m, \ldots, \mathbf{\Phi}^{F_{\mathrm{de}}s-1}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{F}^{F_{\mathrm{de}}s}_m \right], \tag{99}$$

where $\mathbf{F}^{F_{\mathrm{de}}s}_m \triangleq [\mathbf{F}_m, \mathbf{F}_m, \ldots, \mathbf{F}_m]^{\mathrm{T}} \in \mathbb{C}^{N^{\mathrm{p}}_{\mathrm{de}} \times m}$ and $\mathbf{\Phi}^k_{N^{\mathrm{p}}_{\mathrm{de}}} \triangleq \mathrm{diag}\left( \left[ 1, e^{-\bar{\jmath}2\pi \frac{k}{N^{\mathrm{p}}_{\mathrm{de}}}}, \ldots, e^{-\bar{\jmath}2\pi \frac{k(N^{\mathrm{p}}_{\mathrm{de}}-1)}{N^{\mathrm{p}}_{\mathrm{de}}}} \right]^{\mathrm{T}} \right)$. Then using (99), we have

$$\mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} = m[\mathbf{I}^s_m, \mathbf{\Phi}^1_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{I}^s_m, \ldots, \mathbf{\Phi}^{F_{\mathrm{de}}s-1}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{I}^s_m], \tag{100}$$

where $\mathbf{I}^{F_{\mathrm{de}}s}_m \triangleq [\mathbf{I}_m, \mathbf{I}_m, \ldots, \mathbf{I}_m]^{\mathrm{T}} \in \mathbb{C}^{N^{\mathrm{p}}_{\mathrm{de}} \times m}$. Utilizing another permutation matrix $\mathbf{P}_{F_{\mathrm{de}}s,m}$ to gather the non-zero elements in (100) together, it can be rewritten as

$$\mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} = m \mathbf{P}^{\mathrm{T}}_{F_{\mathrm{de}}s,m} \mathbf{D}_{s,m} \mathbf{P}_{F_{\mathrm{de}}s,m}, \tag{101}$$

where $\mathbf{D}_{s,m}$ is defined as

$$\begin{aligned} \mathbf{D}_{s,m} &\triangleq \mathbf{P}_{F_{\mathrm{de}}s,m} \mathbf{F}_{N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{m,F_{\mathrm{de}}s} \mathbf{D}^{\mathrm{H}}_{\mathrm{F}} \mathbf{P}^{\mathrm{T}}_{F_{\mathrm{de}}s,m} \\ &= \mathrm{Bdiag}\left( \mathbf{F}_{F_{\mathrm{de}}s}, \mathbf{F}_{F_{\mathrm{de}}s} \widetilde{\mathbf{\Phi}}^1_{N^{\mathrm{p}}_{\mathrm{de}}}, \ldots, \mathbf{F}_{F_{\mathrm{de}}s} \widetilde{\mathbf{\Phi}}^{m-1}_{N^{\mathrm{p}}_{\mathrm{de}}} \right), \end{aligned} \tag{102}$$

where $\widetilde{\mathbf{\Phi}}^k_{N^{\mathrm{p}}_{\mathrm{de}}} \triangleq \mathbf{I}_{F_{\mathrm{de}}s,N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{\Phi}^k_{N^{\mathrm{p}}_{\mathrm{de}}}$. Finally (97) can be rewritten as

$$\begin{aligned} \mathbf{A}_{n_{\mathrm{c}}} \hat{\mathbf{h}}^t_{n_{\mathrm{c}}} = &\mathbf{V}_{\mathrm{s}} \left( \sum_{u=0}^{U_{\mathrm{c}}-1} \hat{\mathbf{H}}_u (\mathbf{X}^{q_u}_{m,N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{D}_{\mathrm{F}} \mathbf{P}_{m,F_{\mathrm{de}}s})^{\mathrm{T}} \right) \\ &\times (\mathbf{X}^r_{N^{\mathrm{p}}_{\mathrm{v}}} \mathbf{I}_{N^{\mathrm{p}}_{\mathrm{v}},N^{\mathrm{p}}_{\mathrm{de}}} \mathbf{P}^{\mathrm{T}}_{F_{\mathrm{de}}s,m} \mathbf{D}_{s,m} \mathbf{P}_{F_{\mathrm{de}}s,m})^{\mathrm{T}}, \end{aligned} \tag{103}$$

which is the efficient implementation of computing $\mathbf{A}_{n_{\mathrm{c}}} \hat{\mathbf{h}}^t_{n_{\mathrm{c}}}$.

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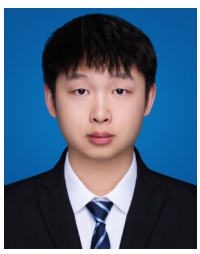
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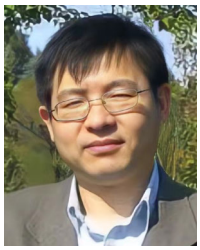
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